Surviving the Future

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Crisis, Climate, and Human Resilience

SETYO WISODO

Independently published · Indonesia

Surviving the Future: Crisis, Climate, and Human Resilience

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First edition, 2026.

ISBN (paperback):
ISBN (ebook):

Data in this book reflect published sources available up to mid-2026. Climate science and statistics are updated regularly; readers are encouraged to consult the primary sources listed in the References.

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Typeset in Source Serif 4 and Source Sans 3 (SIL Open Font License). Cover illustration created for this edition.

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For the communities on every coastline, floodplain and hillside who are already adapting — and for the children who will inherit what we build.

Contents

Preface ix

PART I · UNDERSTANDING THE CRISIS

1 A Warming World: What We Know and How We Know It 3
2 The Pathways Ahead: Scenarios, Tipping Points and Committed Change 9
3 Indonesia and Southeast Asia on the Front Line 15

PART II · THE HAZARDS

4 Heat: The Silent Killer 23
5 Water: Too Much, Too Little 29
6 Food Systems Under Stress 35
7 Health in a Hotter World 41
8 Cascading Risks: Economy, Migration, Conflict and Infrastructure ... 47

PART III · PREPARING HOUSEHOLDS

9 Know Your Risk: A Personal and Local Assessment 55
10 The Resilient Household: Water, Food, Energy and Shelter 61
11 Emergencies: Plans, Kits, Evacuation and First Aid 67
12 Financial Resilience 73

PART IV · STRONGER TOGETHER

13 Community Resilience: Social Capital, Mutual Aid and Local Knowledge 79
14 Cities and Villages That Adapt 85
15 Technology for Resilience: From Solar Panels to Artificial Intelligence 89
16 The Mind in Crisis: Anxiety, Trauma and Psychological Resilience ... 95

PART V · CHANGING THE TRAJECTORY

17 Policy, Finance and Justice 103
18 Thinking Clearly: Evidence, Misinformation and Climate Literacy .. 109
19 A Life Plan for a Changing Century 115
Appendix A: Preparedness Checklists 119
Appendix B: Emergency Contacts and Information Sources (Indonesia) 123

Glossary 127
References 131
About the Author 135
Index 137

Preface

This is a book about living through something that is already happening.

For most of the twentieth century, climate change was discussed in the future tense. Scientists warned that rising concentrations of carbon dioxide would eventually warm the planet, and governments debated whether, when and how to respond. That period is over. The world has already warmed by roughly 1.4 °C compared with the late nineteenth century. The year 2024 was the hottest ever measured, at about 1.55 °C above the pre-industrial average, and the eleven years from 2015 to 2025 are the eleven warmest in a record that stretches back 176 years (WMO, 2026). The question is no longer whether the climate will change during our lifetimes. It is how much it will change, how fast, and how well we will cope.

Three facts make further warming certain. The first is physics: carbon dioxide and other greenhouse gases trap heat, a property measured in laboratories since the 1850s. The second is accumulation: carbon dioxide persists in the atmosphere for centuries, and its concentration has risen from about 278 parts per million before industrialisation to nearly 424 parts per million today (NOAA, 2025a). The third is inertia: the oceans, which have absorbed around 91 percent of the extra energy trapped by greenhouse gases, release that heat slowly, while ice sheets and sea levels respond over centuries (WMO, 2026). Even in the most optimistic scenario, the coming decades will be hotter than the past ones.

None of this means the future is hopeless. Whether the world stabilises at 1.7 °C or drifts towards 3 °C makes an enormous difference to every number in this book — to the frequency of deadly heat, to the height of the sea, to the yield of rice fields and the reach of disease-carrying mosquitoes. And within any level of warming, preparation saves lives. On Simeulue Island, off the coast of Aceh, an oral tradition

about the tsunami of 1907 taught islanders to run to the hills when the ground shook and the sea withdrew. When the far larger tsunami of December 2004 struck, seven people died out of a population of about 78,000 (McAdoo et al., 2006). Knowledge, passed on and acted upon, is a survival tool.

Why this book

There are many excellent books about climate science and many practical manuals about disaster preparedness. Few connect the two, and fewer still are written from the perspective of the tropics — of countries like Indonesia, where the population is large, the coastline is long, the economy is growing and the margin for error is thin. This book tries to fill that gap. It moves from the science of warming, through the specific hazards that warming amplifies, to the concrete steps that households, communities and societies can take.

The book has three commitments. First, every significant factual claim is drawn from a published, checkable source: the assessment reports of the Intergovernmental Panel on Climate Change (IPCC), the World Meteorological Organization (WMO), peer-reviewed journals, and official Indonesian data from agencies such as BMKG, BNPB and the Ministry of Health. Sources are cited in the text by author and year and listed in full at the end. Where the evidence is uncertain, the text says so.

Second, the book is practical. Understanding the greenhouse effect is valuable, but it does not keep a grandmother cool during a heat wave or protect a family's documents during a flood. Each chapter in Parts III and IV ends with actions that can be taken this week, this year and this decade.

Third, the book treats resilience as a collective achievement. Individuals can prepare, but no household is resilient alone. The most robust protection against climate shocks has always been strong communities, competent institutions and fair policies. The Indonesian tradition of *gotong royong* — mutual cooperation — is not a sentimental idea in this context; it is a proven strategy.

How to use this book

The book is organised in five parts.

Part I, Understanding the Crisis, explains how we know the climate is changing, where it is heading, and why Indonesia and Southeast Asia are especially exposed.

Part II, The Hazards, examines heat, water, food, health and the cascading risks that link them to the economy, migration and conflict.

Part III, Preparing Households, is a practical guide to assessing your own risk and preparing your home, family and finances.

Part IV, Stronger Together, looks at communities, cities and villages, technology and mental health.

Part V, Changing the Trajectory, turns to policy, finance and justice, to the skills of thinking clearly in an age of misinformation, and finally to a personal plan for a changing century.

Readers who want the practical material first can begin with Part III and return to the science later. The appendices contain checklists, emergency contacts and a glossary of technical terms. Boxes marked **Key points** summarise each chapter; boxes marked **Indonesia focus** highlight local data and examples; boxes marked **Take action** list concrete steps.

A note on numbers: climate statistics are revised every year as new data arrive. The figures here reflect sources published up to mid-2026. The trends they describe are robust; the exact values will change. Readers are encouraged to check the latest reports, and Chapter 18 explains how.

The future described in this book is demanding. But it is not yet written. What we know, what we prepare, and what we choose to do together will shape it.

Setyo Wisodo Indonesia, 2026

PART I

Understanding the Crisis

CHAPTER 1

A Warming World: What We Know and How We Know It

Climate change is not a matter of belief. It rests on physics understood for more than 150 years, on measurements taken every day across the planet, and on independent lines of evidence that all point in the same direction.

A discovery older than the light bulb

The idea that certain gases warm the Earth is not new. In the 1820s the French mathematician Joseph Fourier reasoned that the planet was warmer than it should be, given its distance from the Sun, and suggested that the atmosphere acted as an insulating layer. In 1856 the American scientist Eunice Foote showed that a glass cylinder filled with carbon dioxide heated more in sunlight than one filled with ordinary air. Three years later, the Irish physicist John Tyndall measured precisely how water vapour and carbon dioxide absorb infrared radiation — the heat that the Earth's surface gives off. And in 1896 the Swedish chemist Svante Arrhenius calculated that doubling the amount of carbon dioxide in the air would warm the planet by several degrees.

These scientists were not activists. They were curious about how the world worked, and the physics they uncovered has been confirmed thousands of times since. Today, the same principles are used to design heat-seeking sensors, weather satellites and the infrared thermometers found in clinics.

How the greenhouse effect works

Sunlight arrives at the Earth mostly as visible light. About 30 percent is reflected back to space by clouds, ice and bright surfaces; the rest is absorbed by the land, the ocean and the atmosphere, warming them. A warm surface emits energy in turn, but at longer, infrared wavelengths. Nitrogen and oxygen, which make up 99 percent of dry air, are transparent to this infrared radiation. Water vapour, carbon dioxide, methane, nitrous oxide and a few other gases are not. They absorb the outgoing heat and re-emit it in all directions, including back towards the surface.

This natural greenhouse effect is essential to life. Without it, the average surface temperature of the Earth would be around −18 °C instead of about 15 °C. The problem is not the existence of the greenhouse effect but its intensification. By burning coal, oil and gas, clearing forests and draining peatlands, humanity has added enough greenhouse gases to thicken the insulating blanket measurably.

KEY GASES AT A GLANCE

- **Carbon dioxide (CO_2)** — from burning fossil fuels, cement production and deforestation. The largest single driver of warming. A substantial fraction of each emission remains in the atmosphere for centuries.
- **Methane (CH_4)** — from livestock, rice paddies, landfills, wetlands and leaks from oil, gas and coal operations. About 80 times more powerful than CO_2 over 20 years, but it breaks down within about a decade.
- **Nitrous oxide (N_2O)** — mainly from nitrogen fertilisers. Long-lived and potent.
- **Water vapour** — the most abundant greenhouse gas, but it acts as an amplifier rather than a driver: warmer air holds more moisture (about 7 percent more per degree Celsius), which traps more heat.

The rise in carbon dioxide

In 1958 the American scientist Charles David Keeling began measuring carbon dioxide at the Mauna Loa Observatory in Hawaii, high above local sources of pollution. His record, now known as the Keeling Curve, is one of the most important data sets in science. It shows a steady sawtooth rise: concentrations dip each northern summer as plants absorb carbon and rise each winter, while the underlying trend climbs relentlessly.

When Keeling began, the concentration was about 315 parts per million (ppm). Before the Industrial Revolution it had been about 278 ppm, a level that ice cores show was never exceeded during the previous 800,000 years. In 2024 the global average reached 422.8 ppm, and the increase during that single year — 3.75 ppm — was the largest ever recorded (NOAA, 2025a). The rise is unambiguously human in origin. Carbon from fossil fuels has a distinctive isotopic signature, and the falling ratio of carbon-13 to carbon-12 in the atmosphere, together with the slow decline in atmospheric oxygen, matches the chemistry of combustion.

The evidence of warming

If greenhouse gases are trapping more heat, the Earth's energy budget should be out of balance, and that extra energy should show up as warming of the air, the ocean and the land, melting of ice and rising seas. All of these changes are observed.

Air temperature. Thermometer records maintained independently by agencies in the United States, the United Kingdom, Japan and Europe agree that the global average surface temperature has risen by more than 1 °C since 1850–1900. The WMO's consolidated analysis puts 2024 at about 1.55 °C above that baseline and 2025 at about 1.43 °C, the second or third warmest year on record (WMO, 2026). Each of the past four decades has been warmer than any decade before it.

Ocean heat. The ocean has absorbed around 91 percent of the excess energy trapped since the 1970s. Ocean heat content reached a new record in 2025, and the rate of ocean warming over the past two decades is more than double that seen earlier (WMO, 2026). Because water expands as it warms, this heat also raises sea level.

Ice. Mountain glaciers are retreating on every continent, including the small tropical glaciers near Puncak Jaya in Papua, which have shrunk dramatically and are expected to disappear entirely. The Greenland and Antarctic ice sheets are losing mass. Arctic sea ice in late summer covers far less area than it did in the 1980s.

Sea level. Global mean sea level rose by about 20 cm between 1901 and 2018 (IPCC, 2021). The rate is accelerating: from 2012 to 2025, it averaged about 4.75 mm per year, nearly twice the rate of 1993–2011 (WMO, 2026).

Extremes. Hot days and nights have become more frequent and cold extremes less frequent across most land areas. Heavy rainfall has intensified in many regions, consistent with a warmer atmosphere holding more moisture.

How we know humans are responsible

Natural factors do influence the climate. The Sun's output varies slightly; volcanic eruptions inject particles that temporarily cool the planet; ocean cycles such as El Niño and La Niña shift heat between the ocean and the atmosphere. Scientists have examined each of these explanations carefully. Solar output has been flat or slightly declining since the 1950s, while temperatures have risen. Volcanoes cause cooling, not warming, and their effects fade within a few years. El Niño and La Niña move heat around but do not create it.

Several fingerprints point specifically to greenhouse gases. The upper atmosphere (the stratosphere) is cooling while the lower atmosphere warms — the pattern expected when heat is trapped near the surface, not when the Sun brightens. Nights are warming faster than days. Satellite measurements show less infrared radiation escaping to space at exactly the wavelengths absorbed by carbon dioxide.

Based on this evidence, the IPCC's Sixth Assessment Report concluded in 2021 that "it is unequivocal that human influence has warmed the atmosphere, ocean and land" (IPCC, 2021). A 2025 update by more than 60 scientists estimated that human activity was responsible for about 1.36 °C of warming in 2024, with the remainder of that year's heat due to natural variability, chiefly El Niño. Human-caused warming is now increasing at about 0.27 °C per decade, a rate unprecedented in the instrumental record (Forster et al., 2025).

SCIENTIFIC CONSENSUS

Surveys of the peer-reviewed literature have repeatedly found overwhelming agreement among climate scientists. A 2021 analysis of more than 88,000 studies found that over 99 percent of papers agreed that recent climate change is mainly caused by humans (Lynas et al., 2021). Every national academy of science of a major country, and every major scientific organisation in the relevant fields, accepts this conclusion.

Attribution: connecting extremes to warming

A newer branch of science, called extreme-event attribution, asks how climate change has altered the likelihood or intensity of a particular heat wave, flood or drought. Researchers compare the real world with simulations of a world without human greenhouse gas emissions. The results are now routine. The IPCC estimates that a hot extreme that occurred once every 50 years in the pre-industrial climate now occurs nearly five times as often, and would occur almost nine times as often at 1.5 °C of warming and about 14 times as often at 2 °C (IPCC, 2021).

Attribution matters because it turns an abstract global average into specific, local consequences. When a heat wave kills hundreds of people, when a monsoon dumps unprecedented rain, it is now possible to say with quantified confidence how much more likely climate change made it.

Certainty and uncertainty

It is important to be honest about what science does not know. Clouds remain a major source of uncertainty: depending on how they respond to warming, they could amplify or dampen it somewhat. The precise behaviour of ice sheets under sustained warming is uncertain, which makes long-term sea-level projections wide. Regional rainfall changes are harder to predict than temperature changes, especially in the tropics.

But uncertainty is not ignorance, and it cuts both ways. The central estimate for how much the planet warms when carbon dioxide doubles — about 3 °C — has been refined over decades, and the IPCC now judges it very unlikely to be below 2 °C (IPCC, 2021). Uncertainty about ice sheets means sea-level rise could be considerably worse than central projections, not only better. A prudent person does not wait for perfect knowledge before buying insurance or fastening a seat belt.

The remaining budget

Because carbon dioxide accumulates, global temperature rises roughly in proportion to the total amount emitted. This gives rise to the idea of a carbon budget: the amount of carbon dioxide that can still be released while keeping warming below a given level. At the start of 2025, the remaining budget for a 50 percent chance of limiting warming to 1.5 °C

was estimated at about 130 billion tonnes of CO_2 — little more than three years of emissions at the current rate of around 42 billion tonnes per year (Forster et al., 2025). In practical terms, the world is very likely to exceed 1.5 °C of long-term warming around 2030.

This does not mean the effort is futile. Every tonne of carbon dioxide avoided reduces future warming. The difference between 1.6 °C and 2.0 °C, or between 2.0 °C and 2.7 °C, is measured in millions of lives and livelihoods. As the WMO's Secretary-General Celeste Saulo put it in late 2025, it is now "virtually impossible" to avoid temporarily overshooting 1.5 °C, but "it's still entirely possible and essential to bring temperatures back down to 1.5 °C by the end of the century" (WMO, 2025a).

KEY POINTS

- The greenhouse effect is basic physics, measured since the 1850s.
- CO_2 has risen from about 278 ppm to nearly 424 ppm, entirely because of human activity.
- The world has warmed by roughly 1.4 °C; 2024 was the hottest year on record at about 1.55 °C.
- Multiple independent fingerprints show the warming is human-caused.
- The 1.5 °C carbon budget is nearly exhausted, but every fraction of a degree avoided matters.

CHAPTER 2

The Pathways Ahead: Scenarios, Tipping Points and Committed Change

The future climate is not a single forecast but a range of possibilities. Which one we live in depends on decisions being made now — though some changes are already locked in.

Why scientists use scenarios

Weather forecasts predict the state of the atmosphere a few days ahead. Climate projections answer a different question: if the world follows a particular path of emissions, land use and development, how will the climate system respond over decades? Because human choices cannot be predicted with physics, scientists use scenarios — internally consistent stories about how population, the economy, technology and policy might evolve.

The IPCC's Sixth Assessment Report uses a family of scenarios called Shared Socioeconomic Pathways (SSPs). Each is labelled with a number describing the socioeconomic story and a number describing the approximate heating effect, in watts per square metre, reached by 2100. Five are used most often:

- **SSP1-1.9** — very rapid emission cuts, net-zero CO_2 around 2050, and removal of carbon dioxide thereafter.
- **SSP1-2.6** — strong mitigation, net-zero CO_2 shortly after 2070.
- **SSP2-4.5** — a "middle of the road" world, roughly in line with current policies.

- **SSP3-7.0** — regional rivalry, slow technological change and rising emissions.
- **SSP5-8.5** — fossil-fuelled development with emissions that roughly double by 2050.

How much warmer?

The table below summarises the IPCC's assessment of global warming in the late twenty-first century under each scenario, relative to 1850–1900 (IPCC, 2021).

Scenario	Best estimate, 2081–2100	Very likely range
SSP1-1.9	1.4 °C	1.0–1.8 °C
SSP1-2.6	1.8 °C	1.3–2.4 °C
SSP2-4.5	2.7 °C	2.1–3.5 °C
SSP3-7.0	3.6 °C	2.8–4.6 °C
SSP5-8.5	4.4 °C	3.3–5.7 °C

Several points stand out. In every scenario, warming continues for at least the next two decades, and the 1.5 °C level is reached or exceeded in the early 2030s or sooner. Only the lowest scenarios bring temperatures back down towards 1.5 °C by 2100, and they depend on removing carbon dioxide from the atmosphere on a large scale. The highest scenario now looks less likely than it did a decade ago, thanks to the falling cost of renewable energy and the climate policies already in place. Independent analyses of current policies suggest a world heading for roughly 2.5–3 °C by 2100 — far better than the worst case, but far from safe.

These numbers can sound small. A day that is 3 °C warmer than yesterday hardly feels like a catastrophe. But global averages hide enormous variation. Land warms faster than the ocean; high latitudes warm faster than the tropics; and the extremes change far more than the average. During the last ice age, about 20,000 years ago, the global average was only around 5–6 °C colder than in pre-industrial times — yet much of North America and Northern Europe lay under ice sheets several kilometres thick, and sea level was about 120 metres lower. A few degrees of global change transform the world.

What changes with each increment

The IPCC emphasises that every increment of warming brings larger and more widespread changes. Compared with 1.5 °C, a world at 2 °C would see:

- more frequent and intense hot extremes over most land areas, with a once-in-fifty-years heat event becoming roughly 14 times as frequent as in the pre-industrial climate, against about 9 times at 1.5 °C;
- heavier rainfall events and more severe droughts in many regions;
- the loss of most tropical coral reefs, which are projected to decline by 70–90 percent at 1.5 °C and by more than 99 percent at 2 °C (IPCC, 2018);
- greater declines in crop yields, fisheries and water availability in vulnerable regions;
- higher sea levels, with the long-term commitment growing substantially.

For a person living in Jakarta, Surabaya or Makassar, these are not abstract differences. They decide how many nights are too hot to sleep without cooling, how often streets flood, whether a fishing family's catch holds steady, and whether the local hospital is overwhelmed with dengue patients.

Sea level: the slow-motion emergency

Of all climate impacts, sea-level rise is the most certain and the most persistent. It is driven by the thermal expansion of warming seawater and by the melting of glaciers and ice sheets. Both respond slowly, which means that sea level will keep rising for centuries even after temperatures stabilise.

By 2100, global mean sea level is projected to rise by about 0.28–0.55 metres relative to 1995–2014 under the lowest scenario and 0.63–1.01 metres under the highest (IPCC, 2021). These are "likely" ranges; the IPCC cannot rule out a rise approaching 2 metres by 2100 under high emissions if ice-sheet processes that are poorly understood kick in. Over the next 2,000 years, sea level could rise by 2–3 metres if warming is limited to 1.5 °C, and by 19–22 metres with 5 °C of warming.

For low-lying coastal areas, the global average is only part of the story. Local factors — especially land subsidence caused by groundwater extraction, as in northern Jakarta and Semarang — can multiply the effective rise many times over. Chapter 3 returns to this.

Tipping points

Most climate changes scale roughly with warming: more heat, more impact. But some parts of the Earth system may respond abruptly once a threshold is crossed, shifting to a new state that is difficult or impossible to reverse on human timescales. These are known as tipping points.

Candidates include the collapse of the Greenland and West Antarctic ice sheets, which together hold enough ice to raise sea level by more than 10 metres over centuries to millennia; the widespread die-off of warm-water coral reefs; abrupt thawing of permafrost, releasing carbon dioxide and methane; the dieback of parts of the Amazon rainforest; and the weakening or collapse of major ocean circulation patterns such as the Atlantic Meridional Overturning Circulation (AMOC).

A 2022 assessment concluded that several tipping points could be triggered within the Paris Agreement range of 1.5–2 °C, and that the risk rises sharply beyond that (Armstrong McKay et al., 2022). The 2025 Global Tipping Points Report went further, judging that warm-water coral reefs have likely already passed their thermal tipping point (Lenton et al., 2025). Since early 2023, the world has experienced its fourth and most extensive global coral bleaching event on record, affecting reefs in more than 80 countries and territories (NOAA, 2025b).

Tipping points carry deep uncertainty: scientists cannot say exactly when they will be crossed, and some may take centuries to play out. But the possibility of irreversible change is a powerful argument for limiting warming as much as possible, and for preparing for outcomes beyond the central projections.

COMMITTED CHANGE VERSUS AVOIDABLE CHANGE

Some impacts are already locked in by past emissions: several decades of further warming, centuries of sea-level rise, the loss of many glaciers and much coral. Others remain avoidable: how hot the late twenty-first century becomes, how fast the seas rise, and whether additional tipping points are crossed. Adaptation is necessary to cope with committed change; mitigation — cutting emissions — is necessary to avoid the worst of what is still avoidable. They are not alternatives.

Overshoot and the long view

Because the 1.5 °C budget is almost spent, many scenarios now involve an “overshoot”: temperatures exceed 1.5 °C for some decades before

being brought back down by reaching net-zero emissions and then removing carbon dioxide from the atmosphere. Overshoot is not harmless. Some damage — species extinctions, ice-sheet loss, sea-level rise — will not reverse when temperatures fall. The longer and higher the overshoot, the greater the irreversible harm.

The WMO estimates an 86 percent chance that at least one year between 2025 and 2029 will exceed 1.5 °C, and a 70 percent chance that the five-year average for that period will do so (WMO, 2025b). Exceeding the threshold in individual years does not mean the Paris goal has formally been breached, which refers to long-term averages. But it signals that the world is entering the zone where the agreement's ambitions become harder to meet.

Living with deep uncertainty

What should an individual or a community do with this information? Three principles help.

Plan for the likely, prepare for the plausible. It is reasonable to plan for a world of roughly 2–3 °C warming by the end of the century, with continued sea-level rise. It is prudent to make sure that critical decisions — where to build a house, what kind of land to buy, where to locate a school or hospital — remain sound if outcomes are worse.

Favour flexible, low-regret actions. Many adaptation measures bring benefits under almost any future: protecting mangroves, improving drainage, building emergency savings, strengthening community networks, improving public health systems. They are worth doing regardless of the precise scenario.

Watch the indicators. Climate risk changes over time. Paying attention to updated assessments, local observations and official warnings allows plans to be adjusted. A plan that is reviewed annually is far more useful than a perfect plan that is never revisited.

KEY POINTS

- Future warming depends on emissions: from about 1.4 °C to over 4 °C by 2100 across IPCC scenarios; current policies point to roughly 2.5–3 °C.
- Every increment matters: the difference between 1.5 °C and 2 °C is large for heat, coral reefs, food and water.

- Sea level will keep rising for centuries; local subsidence can multiply the effect.
- Tipping points may already be in play for coral reefs; others become more likely above 1.5–2 °C.
- Prepare for the likely, protect against the plausible, and choose low-regret actions.

CHAPTER 3

Indonesia and Southeast Asia on the Front Line

An archipelago of more than 17,000 islands, home to about 280 million people and one of the longest coastlines on Earth, sits where the world's warmest ocean meets its most volatile climate patterns.

A country shaped by water

Indonesia's geography makes it unusually exposed to climate change. Its coastline stretches for roughly 99,000 kilometres. A large share of its population, its major cities, its ports, its fish ponds and much of its rice land lie on low coastal plains and river deltas. The country straddles the equator and sits within the Indo-Pacific Warm Pool, the largest body of warm surface water on the planet, which powers the monsoon and the convection that drives weather across Asia and beyond.

Indonesia's climate is also governed by powerful natural cycles. El Niño, in which the eastern Pacific warms and rainfall shifts away from the Maritime Continent, typically brings drought, crop losses and forest fires. La Niña brings the opposite: heavier rain and flooding. The Indian Ocean Dipole, a similar seesaw in the Indian Ocean, can intensify or offset these effects. Climate change does not replace these cycles; it loads the dice, adding warming on top of natural variability and altering how extremes play out.

This exposure combines with vulnerabilities familiar across much of the developing world: rapid urbanisation, informal settlements in hazardous locations, pressure on forests and peatlands, and large numbers of households that depend on farming and fishing and have limited savings to absorb shocks. At the same time, Indonesia has deep insti-

tutional experience with disasters — earthquakes, tsunamis, volcanic eruptions, floods and landslides — and a growing capacity for early warning and disaster management.

Rising temperatures and heat

Average temperatures across Indonesia have risen in line with the global trend, and hot extremes have become more frequent. BMKG, Indonesia's meteorology, climatology and geophysics agency, has recorded a clear warming trend at stations across the archipelago, with the warmest years concentrated in the most recent decade. Because the tropics have a narrow natural temperature range, even modest warming pushes conditions outside the range that people, crops and ecosystems have historically experienced.

The combination of heat and humidity is particularly dangerous. In a humid climate, sweat evaporates less efficiently, so the body struggles to shed heat. Chapter 4 explains the physiology; here it is enough to note that the outdoor workforce — farmers, construction workers, fishers, street vendors, motorcycle couriers — makes up a large share of employment in Indonesia, and is directly exposed.

The sinking coast

The most dramatic climate-related threat to Indonesia's cities is the combination of sea-level rise and land subsidence. Global sea level is rising by about 4–5 millimetres a year. In parts of northern Jakarta, the land itself has historically sunk by several centimetres to more than ten centimetres per year, mainly because of excessive groundwater extraction, compounded by the weight of buildings and the natural compaction of soft sediments (Abidin et al., 2015). Some locations recorded rates of 20–25 centimetres a year in earlier decades. Over the past four decades, parts of North Jakarta have sunk by up to 4 metres, and a significant share of the area now lies below sea level.

There are signs of progress. GPS measurements by the Geological Agency for 2015–2023 suggest that subsidence rates within the Jakarta groundwater basin have slowed to between less than one and about five centimetres per year, compared with 10–20 centimetres in 1997–2005, as groundwater use has been restricted and piped water supply expanded (Badan Geologi, 2025). Satellite radar studies still find rates of up to about 8.5 cm per year along parts of the northern coast (Harintaka

et al., 2024). The lesson is clear: subsidence is a human-made problem with human-made solutions, chiefly replacing groundwater with reliable piped water, but the land that has already sunk will not rise again.

Jakarta is not alone. Semarang, Pekalongan and other cities along the north coast of Java face similar problems, and tidal flooding (*rob*) has become a routine part of life in many coastal neighbourhoods. The decision to move the national capital to Nusantara in East Kalimantan, enacted in Law No. 3 of 2022, reflects many factors, but the vulnerability of Jakarta's coast is part of the context.

INDONESIA FOCUS: THE NORTH COAST OF JAVA

Along the northern coast of Central Java, villages have been abandoned as the sea moves inland. Fish ponds and rice fields have turned brackish, schools and mosques have been raised or relocated, and families have rebuilt floors again and again. Community-led mangrove restoration, permeable sea defences built from bamboo and brushwood that trap sediment, and managed relocation are among the responses being tested. Their success depends on combining engineering with community ownership and on halting the groundwater extraction that drives subsidence.

Water: floods, droughts and saltwater

Indonesia receives abundant rainfall on average, but it arrives unevenly in space and time. Climate change is expected to intensify both ends of the spectrum: heavier downpours during wet seasons and La Niña years, and longer, harsher dry spells during El Niño years and in already drier regions such as Nusa Tenggara.

Urban flooding in Greater Jakarta in early 2020, when record rainfall over New Year's Eve triggered floods that killed dozens of people and forced hundreds of thousands from their homes, illustrated how extreme rain, clogged rivers, inadequate drainage and subsidence combine. In rural areas, flash floods and landslides on deforested slopes are recurring dangers. Meanwhile, droughts threaten rice harvests, drinking water supplies and hydropower, and seawater intrudes into coastal aquifers and rivers as sea level rises and groundwater is overdrawn.

Fires and haze

Indonesia contains roughly 13 million hectares of tropical peatland — waterlogged layers of partly decomposed plant matter, built up over

thousands of years, that store vast amounts of carbon. When peatlands are drained for plantations and agriculture, they dry out and become highly flammable. During El Niño droughts, fires used to clear land can spread uncontrollably, smouldering underground for weeks.

The 2015 fire season was among the worst on record. Around 2.6 million hectares burned between June and October — an area about four and a half times the size of Bali — at an estimated cost to Indonesia of US$16.1 billion, or about 1.9 percent of GDP (World Bank, 2016). Toxic haze blanketed Sumatra, Kalimantan and neighbouring countries for months. One study estimated that the smoke caused around 100,000 premature deaths across Indonesia, Malaysia and Singapore (Koplitz et al., 2016).

Since then, Indonesia has strengthened peatland restoration, fire prevention and law enforcement, and fire seasons in several subsequent years were less severe. But the risk returns with every strong El Niño, and a warmer climate makes droughts more intense.

Oceans, coral reefs and fisheries

Indonesia lies at the heart of the Coral Triangle, the global centre of marine biodiversity. Its reefs support fisheries that provide a major source of animal protein and livelihoods for millions of coastal families, as well as tourism in places such as Raja Ampat, Bali and Wakatobi. Marine heat waves cause coral bleaching; repeated bleaching without time to recover kills reefs. The global bleaching event that began in 2023 has hit Indonesian waters, as it has reefs across the tropics.

Warming seas also push fish towards cooler waters. Studies project that the potential fish catch in tropical waters could decline substantially by mid-century under high emissions, with some of the largest losses in Southeast Asia (Cheung et al., 2010). Ocean acidification, caused by the ocean absorbing carbon dioxide, adds further stress to corals and shellfish.

Health

Rising temperatures and changing rainfall alter the spread of infectious diseases. Dengue fever, transmitted by *Aedes* mosquitoes, is the clearest example. Indonesia recorded its highest number of dengue cases in 2024, with more than 250,000 cases and around 1,400 deaths (Kemenkes, 2025). Warmer temperatures shorten the mosquito's breeding cycle and

the time the virus needs to develop inside it, while irregular rainfall and water storage create breeding sites. Chapter 7 explores the health dimension in detail, including Indonesian innovations such as *Wolbachia*-infected mosquitoes that have sharply reduced dengue in trials in Yogyakarta.

Policy context

Indonesia is both vulnerable to climate change and a significant contributor to it: among the world's larger emitters, with emissions dominated historically by land-use change and peat fires and increasingly by energy, especially coal-fired electricity. Its climate commitments have evolved. In its Enhanced Nationally Determined Contribution (NDC) of 2022, Indonesia pledged to reduce emissions by 31.89 percent by 2030 relative to a business-as-usual scenario on its own, or by 43.20 percent with international support. In its Second NDC, submitted in October 2025, it shifted to absolute targets based on actual 2019 emissions of about 1.145 billion tonnes of CO_2-equivalent, projecting that emissions will peak in 2030 at about 1.34–1.49 billion tonnes before declining towards net-zero emissions by 2060 or sooner (Republic of Indonesia, 2025).

Indonesia is also a recipient of the Just Energy Transition Partnership (JETP), launched at the G20 summit in Bali in 2022 to finance a shift away from coal. Its implementation has been slower than hoped, a subject Chapter 17 returns to. On adaptation, the national development plans and BNPB's disaster risk management framework increasingly integrate climate risk, and programmes such as *Desa Tangguh Bencana* (Disaster Resilient Villages) and *Program Kampung Iklim* (Climate Village Programme) work at the community level.

Why the regional view matters

Indonesia's neighbours face similar threats. The Philippines endures some of the world's strongest typhoons. Vietnam's Mekong Delta, which produces much of the country's rice, is sinking and salinising. Bangkok and Manila face subsidence and flooding comparable to Jakarta's. Myanmar, Bangladesh and the Pacific islands are among the most exposed places on Earth. The region's economies are linked by trade in rice, fish, palm oil and manufactured goods, and by migration. A

drought in one country can raise food prices in another; haze from one country's fires fills the lungs of another's citizens.

This shared exposure is also an opportunity. ASEAN countries cooperate on disaster management through the AHA Centre, share weather data and early warnings, and increasingly coordinate on haze and energy. Regional cooperation will be one of the most important tools for surviving the decades ahead.

KEY POINTS

- Indonesia's long coastline, low-lying cities, peatlands, reefs and climate-sensitive livelihoods make it highly exposed.
- Land subsidence from groundwater extraction multiplies sea-level rise in northern Java; it can be slowed by replacing groundwater with piped water.
- El Niño years bring drought and fire risk; the 2015 fires cost US$16.1 billion and caused an estimated 100,000 premature deaths regionally.
- Dengue reached record levels in 2024.
- Indonesia's Second NDC targets an emissions peak in 2030 and net zero by 2060 or sooner.

PART II

The Hazards

CHAPTER 4

Heat: The Silent Killer

Heat kills more people than any other weather hazard, yet its victims are often uncounted. They die quietly, at home, from strain on the heart and kidneys, days after the temperature peaks.

An underestimated danger

Floods and storms produce dramatic images. Heat does not. There is no destroyed bridge, no collapsed house — only people who feel unwell, become confused, and in the worst cases die, often alone and indoors. Death certificates rarely list "heat" as the cause; they record heart failure, stroke, kidney failure or respiratory disease, conditions that heat has triggered or worsened. Only by comparing deaths during hot spells with deaths expected for the time of year do researchers see the full toll.

That toll is large and growing. The 2025 report of the *Lancet* Countdown on health and climate change estimated that heat-related deaths worldwide averaged about 546,000 per year in 2012–2021, and that the rate of heat-related mortality has risen by 23 percent since the 1990s. Heat exposure also cost an estimated 639 billion potential hours of work in 2024, equivalent to income losses of about US$1.09 trillion (Romanello et al., 2025).

How the body handles heat

The human body keeps its core temperature close to 37 °C. When it gains heat — from the environment or from physical work — it sheds it in two main ways. Blood vessels near the skin widen, carrying heat to the surface where it can radiate away, and sweat glands release moisture that cools the skin as it evaporates. Both mechanisms have costs. Widened

blood vessels make the heart pump harder. Sweating loses water and salts that must be replaced.

Problems arise when the environment makes heat loss difficult. When the air is hotter than the skin (around 35 °C), the body gains heat from its surroundings instead of losing it, and evaporation of sweat becomes the only cooling mechanism. When the air is also very humid, sweat evaporates slowly and drips off without cooling. This is why humid heat, common across Indonesia, is especially dangerous.

Measuring dangerous heat

Air temperature alone is a poor guide to heat stress. Several measures combine temperature with humidity and other factors:

- **Heat index** ("feels like" temperature) combines air temperature and relative humidity.
- **Wet-bulb temperature** is the lowest temperature to which air can be cooled by evaporating water. It indicates how effectively sweat can cool the body.
- **Wet-bulb globe temperature (WBGT)** also accounts for sunlight and wind. It is used to set safe work and exercise limits.

A wet-bulb temperature of 35 °C has been proposed as the theoretical upper limit of human survival, because at that point even a healthy person at rest in the shade cannot shed heat (Sherwood & Huber, 2010). Laboratory studies suggest the practical limit is lower: around 30–31 °C wet-bulb for young, healthy adults in humid conditions (Vecellio et al., 2022), and lower still for older people. Such conditions are still rare, but they are being approached more often in parts of South Asia, the Persian Gulf and, during extreme events, Southeast Asia.

The spectrum of heat illness

Heat illness ranges from mild to fatal. Recognising the stages saves lives.

Heat rash and cramps. Small red bumps where sweat ducts are blocked; painful muscle spasms caused by salt loss. Rest, cool down and drink fluids with electrolytes.

Heat exhaustion. Heavy sweating, weakness, dizziness, headache, nausea, a fast weak pulse, cool clammy skin, and sometimes fainting. Move the person to a cool place, loosen clothing, cool them with water and fanning, and give water or oral rehydration solution if they can

drink. If symptoms worsen or do not improve within an hour, seek medical help.

Heat stroke. A medical emergency. Body temperature above 40 °C, confusion, slurred speech, agitation, seizures or loss of consciousness. Skin may be hot and dry or still sweating. Call emergency services (112 or 119 in Indonesia) immediately and start cooling at once — ideally by immersing the body in cool or cold water up to the neck, or by soaking with water, applying ice packs to the neck, armpits and groin, and fanning continuously. Do not give fluids to a confused or unconscious person. Rapid cooling within the first 30 minutes is the most important factor in survival.

WHO IS MOST AT RISK?

- Older people, especially those living alone
- Infants and young children
- Pregnant women
- People with heart, lung or kidney disease, diabetes or obesity
- People taking certain medications (for example diuretics, some blood-pressure and psychiatric drugs)
- Outdoor workers and athletes
- People in poorly ventilated housing, informal settlements or top-floor rooms under metal roofs
- People experiencing homelessness

Heat in the city

Cities are hotter than the surrounding countryside — often by several degrees, especially at night. This urban heat island effect arises because asphalt, concrete and dark roofs absorb sunlight and release it slowly; buildings trap heat and block wind; vehicles, air conditioners and industry add waste heat; and there is little vegetation or open water to provide evaporative cooling.

Hot nights are especially harmful. When temperatures stay high after dark, the body cannot recover from the day's heat stress, and sleep suffers. In dense neighbourhoods with small, poorly ventilated houses and metal roofs, indoor temperatures can exceed outdoor temperatures well into the night.

The solutions are known: more trees and green spaces, reflective or green roofs, lighter pavements, better ventilation, shade over walkways and markets, and building designs that use cross-ventilation and shad-

ing. Traditional Indonesian architecture — raised floors, steep roofs with wide overhangs, open walls and natural materials — embodied much of this knowledge.

Heat and work

In much of Indonesia and Southeast Asia, a large share of workers labour outdoors or in hot indoor environments such as factories and kitchens. As temperatures rise, they must either slow down, take more breaks or risk illness. The International Labour Organization projects that heat stress will cause productivity losses equivalent to tens of millions of full-time jobs globally by 2030, with South and Southeast Asia most affected.

Simple measures make a large difference: scheduling heavy work in the cooler morning and late afternoon; providing shade, cool drinking water and rest breaks; allowing workers to acclimatise gradually; training supervisors to recognise heat illness; and adjusting quotas on hot days. Over the longer term, mechanisation, better building design and labour standards that set heat limits are needed.

Heat action plans

Many cities and countries have developed heat action plans that combine early warning, public communication, health-system preparation and targeted support for vulnerable people. The city of Ahmedabad in India launched one of the first in South Asia in 2013, after a severe heat wave in 2010; studies found that heat-related mortality fell after the plan was implemented.

A good heat action plan includes:

1. **Forecasting and thresholds** — using weather forecasts to issue graded alerts when heat is expected to reach dangerous levels.
2. **Communication** — clear messages via television, radio, social media, mosques, schools and community leaders about how to stay safe.
3. **Health system readiness** — training health workers, stocking oral rehydration salts and cooling equipment, and preparing for surges in patients.
4. **Protection of vulnerable groups** — checking on older people living alone, opening cooling centres, adjusting school and work hours.
5. **Long-term measures** — urban greening, cool roofs, better housing.

BMKG publishes temperature forecasts and warnings, and local health offices can build on them. Every community can adapt the same principles at a smaller scale.

TAKE ACTION: STAYING SAFE IN EXTREME HEAT

- Drink water regularly, before you feel thirsty. Limit alcohol and very sugary drinks.
- Stay out of direct sun between about 11:00 and 15:00. Wear light, loose, light-coloured clothing and a hat.
- Keep your home cool: close curtains on sun-facing windows during the day, open windows at night when it is cooler, use fans, and spend the hottest hours in the coolest room.
- A damp cloth on the neck, a cool shower or soaking your feet in water helps lower body temperature.
- Never leave children, older people or pets in a parked vehicle, even for a few minutes.
- Check on older relatives and neighbours at least once a day during heat waves.
- Know the signs of heat stroke — confusion, very high temperature, collapse — and call 112 or 119.

KEY POINTS

- Heat is the deadliest weather hazard; about 546,000 people die from heat each year globally.
- Humid heat is especially dangerous because sweat cannot evaporate efficiently.
- Heat stroke is an emergency: cool first, fast, and call for help.
- Cities trap heat; trees, shade, ventilation and cool roofs reduce it.
- Heat action plans and simple workplace measures save lives and livelihoods.

CHAPTER 5

Water: Too Much, Too Little

Most climate impacts reach people through water: through floods that destroy homes, droughts that empty wells, storms that surge ashore and salt that creeps into fields and aquifers.

A more intense water cycle

The physics is simple. Warmer air can hold about 7 percent more water vapour for every degree Celsius of warming. More moisture in the atmosphere means that when conditions favour rain, more of it can fall in a short time. At the same time, higher temperatures increase evaporation from soils, plants and reservoirs, drying the land faster between rains. The result is an intensified water cycle: wetter wet periods, drier dry periods, and more abrupt swings between them.

The IPCC concludes that heavy precipitation has become more frequent and intense over most land regions with good observational coverage, and that this trend will continue with further warming (IPCC, 2021). Agricultural and ecological droughts — periods when soils are too dry for crops and plants — have increased in some regions and will become more severe in many others.

Floods

Floods come in several forms, each with different warning times and dangers.

River floods occur when prolonged rain fills river channels beyond capacity. They usually develop over days, allowing time for warnings, but can affect very large areas.

Flash floods develop within hours, or even minutes, of intense rain, especially in steep catchments and urban areas with sealed surfaces. They are the deadliest type because they leave little time to react.

Urban (pluvial) floods occur when rainfall overwhelms drainage systems, regardless of river levels. They are increasingly common as cities expand over natural wetlands and green spaces.

Coastal floods result from high tides, storm surges and waves, increasingly aggravated by sea-level rise and subsidence. Tidal flooding along Java's north coast is a daily reality in some neighbourhoods.

Floodwater is dangerous in ways people often underestimate. Just 15 centimetres of fast-moving water can knock an adult off their feet; 60 centimetres can carry away a car. Floodwater may hide open drains, debris and downed power lines, and is often contaminated with sewage, chemicals and disease-causing organisms such as the bacteria that cause leptospirosis.

TAKE ACTION: DURING A FLOOD

- Follow official warnings and evacuation orders from BPBD, BNPB and BMKG.
- Move people, documents and valuables to higher floors or higher ground early.
- Switch off electricity at the main switch if water is likely to enter the house — but never touch electrical equipment while standing in water.
- Never walk, swim or drive through floodwater. "Turn around, don't drown."
- Keep children away from floodwater and drains.
- After the flood, wear boots and gloves when cleaning; disinfect surfaces; discard food that has touched floodwater; and watch for fever, muscle pain or jaundice, which may signal leptospirosis.

Landslides

In mountainous and hilly regions, intense rain saturates soils and triggers landslides, especially on slopes stripped of forest, cut for roads, or loaded with houses and terraces. Indonesia experiences hundreds of landslides each year. Warning signs include new cracks in the ground or walls, tilting trees or posts, doors and windows that suddenly jam, springs that appear or become muddy, and rumbling sounds from the slope. Communities on steep slopes should identify safe areas and

evacuation routes, monitor rainfall and warning signs during the rainy season, and avoid building at the foot or crest of steep slopes.

Drought and water scarcity

Drought is a slow-onset disaster. It begins with below-normal rainfall (meteorological drought), then dries the soil (agricultural drought), lowers rivers, reservoirs and groundwater (hydrological drought), and finally affects incomes, food supply and health (socioeconomic drought). Because it develops gradually, its impact is easy to underestimate until wells run dry and crops fail.

In Indonesia, droughts are closely linked to El Niño and positive phases of the Indian Ocean Dipole. During these years, the dry season starts earlier, lasts longer and is more intense, especially in Java, Bali, Nusa Tenggara and southern parts of Sumatra, Kalimantan and Sulawesi. Village wells dry up, women and children spend more time collecting water, farmers delay planting, and the risk of fires rises.

Water scarcity is not only about rainfall. It also reflects how water is managed: leaking pipes, polluted rivers, deforested catchments, overextracted aquifers and unequal access. In many places, improving water management offers greater gains than any new source of supply.

Salt in the water

As sea levels rise and groundwater is pumped from coastal aquifers, seawater moves inland underground. Coastal wells become brackish, rice fields and fish ponds become too salty, and drinking water must be bought or trucked in. River deltas face saltwater pushing further upstream during dry seasons. Salinity is a quiet but serious threat to coastal food production and health — high salt intake from drinking water has been linked to high blood pressure, including in pregnant women.

Responses include stopping excessive groundwater extraction, harvesting rainwater, planting salt-tolerant rice varieties, shifting from rice to aquaculture or salt-tolerant crops in affected areas, and protecting mangroves and coastal wetlands that buffer the land from the sea.

Storms and storm surges

Indonesia lies mostly near the equator, where tropical cyclones rarely form because the Earth's rotation is too weak to spin them up. However, cyclones do affect the southern parts of the archipelago. In April 2021, Tropical Cyclone Seroja caused devastating floods and landslides in East Nusa Tenggara, killing more than 180 people. Cyclones passing near Indonesia can also bring heavy rain, high waves and strong winds far from their centres. Globally, the proportion of tropical cyclones reaching the most intense categories is projected to increase with warming, and rainfall rates within them will rise (IPCC, 2021). Combined with higher sea levels, storm surges will reach further inland.

Securing water at home and in the community

Water security begins at home and in the neighbourhood. Practical measures include:

- **Rainwater harvesting** — collecting runoff from roofs in clean, covered tanks provides a backup supply during dry spells and reduces flooding. Tanks must be covered to prevent mosquito breeding.
- **Infiltration wells and biopores** — small holes or wells that allow rainwater to soak into the ground recharge groundwater and reduce runoff. Many Indonesian cities promote *sumur resapan* and *lubang biopori*.
- **Water storage for emergencies** — keeping enough safe drinking water for at least three days, ideally longer (see Chapter 10).
- **Household water treatment** — boiling, filtering or chlorinating water when supply is uncertain.
- **Protecting springs and catchments** — community forest and spring protection, as practised in many Indonesian villages, preserves water at its source.
- **Reducing waste** — fixing leaks and reusing greywater for gardens.

At the city scale, the key measures are expanding reliable piped water to stop groundwater extraction, restoring rivers and retention ponds, keeping drains free of waste, preserving green space and wetlands, and planning development away from floodplains.

> **INDONESIA FOCUS: THE SPONGE CITY IDEA**
> Instead of relying only on concrete channels to push water away as fast as possible, "sponge city" approaches — pioneered at scale in China — use

parks, wetlands, permeable pavements, green roofs and retention ponds to absorb, store and slowly release rainwater. Elements of this approach are visible in Indonesian programmes for retention ponds (*waduk* and *embung*), river normalisation combined with green space, and infiltration wells. The approach works best when combined with strict control of building on floodplains and riverbanks.

KEY POINTS

- Warmer air holds more moisture, intensifying both heavy rain and drought.
- Flash floods are the deadliest flood type; never enter floodwater.
- Drought develops slowly; El Niño years bring higher risk in much of Indonesia.
- Sea-level rise and groundwater pumping push salt into coastal water and soils.
- Rainwater harvesting, infiltration, storage and good management build water security.

CHAPTER 6

Food Systems Under Stress

Climate change threatens food at every step — in the field, on the boat, in storage, on the road and at the market — and it hits hardest those who already spend most of their income on food.

Food security in four dimensions

Food security exists, in the definition used by the Food and Agriculture Organization (FAO), when all people at all times have physical and economic access to sufficient, safe and nutritious food. It has four dimensions: **availability** (is enough food produced?), **access** (can people afford it and reach it?), **utilisation** (is it safe and nutritious, and can the body use it?), and **stability** (are the other three reliable over time?). Climate change affects all four.

Crops in a warmer world

Plants are sensitive to temperature at critical stages. High temperatures during flowering can sterilise pollen; hot nights increase respiration, burning energy that would otherwise fill grains; drought stunts growth; floods drown seedlings; and new pest and disease patterns emerge. A large synthesis of studies estimated that, without effective adaptation, each degree Celsius of global warming reduces average global yields of maize by about 7.4 percent, wheat by 6.0 percent, rice by 3.2 percent and soybeans by 3.1 percent (Zhao et al., 2017).

Rice deserves special attention in Indonesia, where it is the staple food and the basis of rural livelihoods. Research at the International Rice Research Institute found that rice yields fell by about 10 percent for each 1 °C increase in night-time temperature during the dry season (Peng et al., 2004). Rice is also vulnerable to drought, to flooding and

to saltwater intrusion in coastal areas. At the same time, flooded rice paddies are a significant source of methane, so rice systems sit at the centre of both adaptation and mitigation.

Carbon dioxide itself has mixed effects. Higher concentrations can boost photosynthesis in some crops, but experiments show that rice and wheat grown under elevated carbon dioxide contain lower concentrations of protein, zinc and iron — a hidden threat to nutrition for populations that depend on these staples.

Fisheries and aquaculture

Fish are an essential source of protein and micronutrients for Indonesians, and the fisheries sector supports millions of livelihoods. Warming waters, marine heat waves, coral bleaching, ocean acidification and declining oxygen in parts of the ocean are reshaping marine life. Many fish species are shifting towards cooler, deeper or more distant waters, and tropical regions are projected to see some of the largest declines in potential catch (Cheung et al., 2010). Small-scale fishers with small boats cannot easily follow fish further offshore, and rougher seas reduce the number of days they can safely work.

Aquaculture — fish, shrimp and seaweed farming — has grown rapidly in Indonesia and offers opportunities, but coastal ponds are exposed to flooding, salinity changes and disease outbreaks linked to temperature. Seaweed farms, important for many coastal communities, have suffered from heat-related disease ("ice-ice") during warm periods.

Price shocks and access

For poor households, the most immediate impact of climate change on food is often price. When droughts or floods damage harvests in a major producing region, prices rise locally and sometimes globally. Rice, cooking oil, chillies, shallots and other staples in Indonesia are sensitive to weather, and price spikes in these commodities are felt immediately in household budgets. Poor families spend a much larger share of their income on food, so a price increase that is an inconvenience for a middle-class household can force a poor family to eat less, switch to cheaper and less nutritious foods, sell assets, or take children out of school.

Global food markets connect distant weather events. In 2007–2008 and again in 2010–2011, a combination of droughts in exporting coun-

tries, export restrictions, high energy prices and speculation drove sharp rises in world food prices, contributing to unrest in several countries. As climate extremes become more frequent, the chance of simultaneous harvest failures in multiple breadbaskets rises.

Nutrition and health

Food insecurity is not only about calories. Indonesia still faces a significant burden of childhood stunting — children who are too short for their age because of chronic undernutrition — alongside rising obesity and diet-related disease. Climate shocks can worsen both by reducing access to diverse, nutritious foods and pushing families towards cheap, energy-dense processed foods. Heat can also spoil food faster, increasing the risk of food poisoning, especially where refrigeration is limited.

Adapting farming

Farmers have always adapted to variable weather. The challenge now is to adapt faster and further. Proven approaches include:

- **Improved varieties** — rice and other crops bred for tolerance to heat, drought, flooding (for example "submergence-tolerant" rice that survives two weeks under water) and salinity.
- **Water management** — small reservoirs (*embung*), efficient irrigation, and alternate wetting and drying in rice paddies, which saves water and reduces methane emissions.
- **Adjusted planting calendars** — using seasonal forecasts from BMKG to time planting. Indonesia's *Kalender Tanam* (planting calendar) system and climate field schools (*Sekolah Lapang Iklim*) help farmers interpret forecasts.
- **Diversification** — growing a mix of crops, integrating trees, livestock and fish ponds (as in traditional *mina padi* rice-fish systems), and diversifying income sources.
- **Soil health** — organic matter, cover crops and reduced burning improve the soil's ability to hold water.
- **Crop insurance** — Indonesia's subsidised rice farm insurance (*Asuransi Usaha Tani Padi*) compensates farmers for losses from floods, droughts and pests, though uptake remains limited.
- **Post-harvest improvements** — better drying, storage and cold chains reduce losses that grow as heat and humidity rise.

Food systems and emissions

Food systems — including land clearing, agriculture, processing, transport and waste — account for roughly a third of global greenhouse gas emissions. In Indonesia, the conversion of forests and peatlands for agriculture has been a major source of emissions. Reducing food loss and waste, protecting forests and peatlands, improving fertiliser use, managing rice paddy water and shifting diets towards more plant-based foods where appropriate can cut emissions while also improving resilience.

INDONESIA FOCUS: LOCAL FOOD DIVERSITY

Before rice became dominant almost everywhere, many Indonesian regions relied on diverse staples: sago in Maluku and Papua, maize in Madura and Nusa Tenggara, cassava, sweet potato, sorghum and taro elsewhere. Many of these crops are more tolerant of drought or poor soils than rice. Renewed interest in local food diversity (*diversifikasi pangan*) is not nostalgia; it is a strategy for reducing dependence on a single climate-sensitive crop.

What households can do

Households cannot control global food markets, but they can reduce their exposure. Keeping a modest stock of non-perishable staples (see Chapter 10), growing vegetables, herbs or fruit trees where space allows, joining or supporting local farmers' groups and cooperatives, reducing food waste, and learning to cook with a variety of staples all increase resilience. At the community level, village food barns (*lumbung pangan*), cooperative purchasing, and links between urban consumers and nearby farmers can buffer price shocks.

KEY POINTS

- Each degree of warming reduces average global yields of major crops, including rice.
- Hot nights, drought, flooding and salinity are key threats to Indonesian rice.
- Fisheries are shifting as seas warm; small-scale fishers are especially vulnerable.
- Price shocks hit poor households hardest and can worsen nutrition.

- Improved varieties, water management, forecasts, diversification and insurance help farmers adapt.

CHAPTER 7

Health in a Hotter World

The World Health Organization has called climate change the single biggest health threat facing humanity. It acts through heat, disease, air, water, food and the mind — and it strains the very systems meant to protect us.

Many pathways, one body

Climate change affects health directly, through heat, floods and storms, and indirectly, through changes in infectious diseases, air quality, water and food security, displacement and mental health. These pathways interact. A heat wave raises demand on hospitals at the same time as it causes power cuts; a flood spreads waterborne disease while damaging clinics and roads; a drought reduces food intake and forces families to drink unsafe water.

The *Lancet* Countdown, an annual assessment by more than 100 experts, tracks dozens of indicators of how climate change affects health. Its 2025 report found that many indicators had reached record levels, including heat-related deaths, labour losses and the climatic suitability for dengue transmission (Romanello et al., 2025).

Dengue and other mosquito-borne diseases

Dengue is the fastest-spreading mosquito-borne viral disease in the world, and Indonesia carries one of the heaviest burdens. The virus is transmitted mainly by the *Aedes aegypti* mosquito, which breeds in clean standing water in and around homes — in water storage jars, flower vases, discarded tyres, gutters and plastic waste. Warmer temperatures speed up the mosquito's life cycle and shorten the time the virus needs to replicate inside it, making transmission more efficient. Rainfall cre-

ates breeding sites, but so does drought, when households store water in open containers.

Indonesia recorded its highest-ever number of dengue cases in 2024: more than 250,000 cases and around 1,400 deaths (Kemenkes, 2025). Most dengue infections cause mild illness or none at all, but a small proportion progress to severe dengue, with bleeding, plasma leakage and shock. Warning signs — severe abdominal pain, persistent vomiting, bleeding from the gums or nose, blood in vomit or stool, restlessness or lethargy — usually appear as the fever falls, around days three to seven. Anyone with these signs needs urgent medical care.

INDONESIA FOCUS: WOLBACHIA MOSQUITOES

In Yogyakarta, researchers from Universitas Gadjah Mada and the World Mosquito Program tested a novel approach: releasing *Aedes aegypti* mosquitoes carrying *Wolbachia*, a naturally occurring bacterium that blocks the dengue virus from replicating. In a randomised controlled trial, areas receiving *Wolbachia* mosquitoes saw a 77 percent reduction in dengue cases and an 86 percent reduction in dengue hospitalisations (Utarini et al., 2021). The approach has since been expanded to other Indonesian cities. It complements, rather than replaces, the classic *3M Plus* approach: *menguras* (drain), *menutup* (cover) and *mendaur ulang* (recycle) water containers, plus measures such as repellents and larvicides.

Malaria, once widespread in Indonesia, has been eliminated from many districts but remains endemic in parts of eastern Indonesia, particularly Papua. Changing temperature and rainfall patterns can shift its range, including to higher elevations. Chikungunya and Zika, also spread by *Aedes* mosquitoes, may follow similar patterns to dengue.

Waterborne and foodborne diseases

Floods contaminate drinking water with sewage; droughts concentrate pathogens and push people towards unsafe sources; heat accelerates the growth of bacteria in food and water. Diarrhoeal diseases, including cholera and typhoid, rise after floods and during hot periods, especially where sanitation is poor. Leptospirosis, a bacterial infection spread through water contaminated by the urine of rats and other animals, often surges after floods in Indonesian cities.

Basic protection is effective: safe drinking water (boiled, filtered or treated), handwashing with soap, safe food storage, and proper sanita-

tion. Oral rehydration solution — clean water with the right balance of salt and sugar — prevents most deaths from diarrhoea. The WHO's home recipe is one litre of clean water, six level teaspoons of sugar and half a level teaspoon of salt; commercial oral rehydration salts (*oralit*) are better where available.

Air: smoke, haze and pollution

Climate change and air pollution are closely linked. Burning fossil fuels causes both. Heat speeds up the formation of ground-level ozone. Droughts increase the risk of forest and peatland fires, whose smoke contains fine particles (PM2.5) that penetrate deep into the lungs and bloodstream. Haze episodes in Indonesia have caused surges in respiratory infections, asthma attacks and cardiovascular problems, and long-term exposure to fine particles shortens lives. The *Lancet* Countdown estimated that fine-particle pollution from wildfire smoke caused a record 154,000 deaths worldwide in 2024 (Romanello et al., 2025).

During haze or heavy pollution, stay indoors with windows closed where possible, reduce physical exertion, use a properly fitted N95 or equivalent mask when outdoors (ordinary cloth or surgical masks offer little protection against fine particles), and seek medical advice if you have asthma, heart or lung disease. Air quality information is available from BMKG and the Ministry of Environment.

Heat and chronic disease

Chapter 4 covered heat illness. Beyond acute heat stroke, heat worsens many chronic conditions. It strains the heart, raises the risk of heart attacks and strokes, worsens kidney disease (repeated dehydration damages the kidneys, and chronic kidney disease among agricultural workers in hot regions is a growing concern), and is associated with adverse pregnancy outcomes, including preterm birth. Heat also affects mental health, with studies linking high temperatures to increases in emergency visits for mental health conditions, aggression and suicide.

Mental health

Disasters cause trauma, grief and loss. People who lose homes, livelihoods or loved ones in floods, fires or storms face elevated risks of depression, anxiety and post-traumatic stress disorder. Slow-onset

changes — repeated crop failures, loss of land to the sea, the disappearance of familiar landscapes — cause chronic stress and a form of grief sometimes called *solastalgia*: distress at the degradation of one's home environment. And many people, especially the young, experience anxiety about the future of the planet. Chapter 16 addresses these issues and how to cope with them.

Health systems under strain

Hospitals and clinics are themselves vulnerable. Floods can inundate ground floors, damage equipment and cut access; heat can overwhelm cooling systems and spoil medicines and vaccines; power cuts disable equipment; and surges in patients during heat waves, epidemics or disasters stretch staff thin. Climate-resilient health systems need robust infrastructure (raised generators, backup power and water, flood-proof designs), surveillance systems that link climate data with disease data to predict outbreaks, trained staff, stockpiles of essential supplies, and strong primary care that reaches communities before crises escalate.

The health sector is also a source of emissions — globally around 4–5 percent of the total — and can lead by example through energy efficiency, renewable power and reducing waste.

TAKE ACTION: PROTECTING YOUR FAMILY'S HEALTH

- Practise *3M Plus* weekly: drain, cover and recycle water containers; use repellent and window screens.
- Know the warning signs of severe dengue and seek care immediately if they appear.
- Keep oral rehydration salts, a thermometer and basic medicines at home.
- During haze, stay indoors when possible and use N95 masks outdoors.
- After floods, avoid wading in water, use boots and gloves for cleaning, and watch for fever and muscle pain.
- Keep vaccinations up to date and keep copies of medical records in your emergency documents.

KEY POINTS

- Climate change affects health through heat, infectious disease, air, water, food and mental health.
- Dengue reached record levels in Indonesia in 2024; *Wolbachia* mosquitoes cut cases by 77 percent in a Yogyakarta trial.

- Floods bring diarrhoeal disease and leptospirosis; safe water and hygiene are the best defences.
- Wildfire smoke and pollution harm lungs and hearts; N95 masks protect against fine particles.
- Health systems must be strengthened to withstand climate shocks.

CHAPTER 8

Cascading Risks: Economy, Migration, Conflict and Infrastructure

Climate shocks rarely stay in one place or one sector. A drought becomes a price spike, a power cut, a migration, a political crisis. Understanding how risks cascade is the key to stopping them.

When risks connect

The IPCC describes climate risks as increasingly "complex and compound": multiple hazards occur at the same time or in sequence, and their impacts cascade across sectors and borders (IPCC, 2022). A heat wave during a drought increases fire risk and electricity demand while reducing hydropower and cooling water for thermal power plants. A flood that cuts a highway disrupts supply chains far from the flooded area. A harvest failure in one exporting country raises food prices around the world.

These connections mean that the total damage from climate change can be much larger than the sum of individual impacts. They also mean that resilience in one system — a robust power grid, a diversified food supply, a strong social safety net — can protect others.

The economic toll

Climate-related disasters already impose large and rising economic costs. Insured losses from natural catastrophes worldwide have exceeded US$100 billion in each recent year, and total economic losses — including uninsured damage — are considerably higher. In developing

countries, most losses are uninsured, falling on households, businesses and governments.

Beyond the direct damage, climate change acts as a drag on economic growth. Heat reduces labour productivity, especially in agriculture, construction and manufacturing. Damage to infrastructure diverts public budgets from development to reconstruction. Uncertainty discourages investment. Studies of historical data suggest that unusually hot years reduce economic growth in already hot countries, and that the effects can persist.

For Indonesia, the costs are visible in recurring flood damage in cities, losses from drought and fire, coastal erosion and tidal flooding, and the costs of disease. The 2015 fires alone cost an estimated US$16.1 billion (World Bank, 2016). Investments in adaptation — drainage, coastal protection, early warning, resilient infrastructure — are expensive, but the evidence consistently shows that they cost far less than the damage they prevent.

> **THE ECONOMICS OF PREVENTION**
> The Global Commission on Adaptation estimated that investing US$1.8 trillion globally between 2020 and 2030 in five areas — early warning systems, climate-resilient infrastructure, improved dryland agriculture, mangrove protection, and resilient water resources — could generate US$7.1 trillion in total net benefits. It also found that just 24 hours' warning of a coming storm or heat wave can cut the ensuing damage by 30 percent (Global Commission on Adaptation, 2019).

Displacement and migration

People have always moved in response to environmental change. Today, climate change is intensifying the pressures that drive movement, while also making it harder for the poorest to move at all.

Sudden disasters cause large-scale displacement. The Internal Displacement Monitoring Centre records tens of millions of new internal displacements caused by disasters every year — the large majority from weather-related events such as floods and storms. Most people displaced by disasters return home within weeks or months, but some cannot, and repeated displacement erodes livelihoods and savings.

Slow-onset changes — sea-level rise, salinisation, drought and declining agricultural productivity — drive more gradual migration, typically

from rural areas to cities. The World Bank's *Groundswell* report projected that, without strong action, climate change could force up to 216 million people across six world regions to move within their own countries by 2050, including up to 49 million in East Asia and the Pacific (Clement et al., 2021).

Migration is not only a problem. It can be an effective adaptation strategy, allowing households to diversify income through remittances and to move away from danger. The challenges arise when migration is forced, unplanned and unsupported — when people arrive in cities without jobs, housing or services, often settling in informal neighbourhoods that are themselves exposed to floods and heat. Policies that support safe, orderly and voluntary movement, including well-planned relocation where staying is no longer viable, can turn a source of vulnerability into a source of resilience.

Conflict and security

Climate change does not cause wars by itself, and the relationship between climate and conflict is debated among researchers. But there is broad agreement that climate change acts as a "threat multiplier": it can intensify existing tensions over land, water and resources, strain governments, undermine livelihoods and increase the pool of people with grievances, particularly where institutions are weak and inequality is high. Disputes between farmers and herders, competition over water, and tensions arising from displacement have been linked to climate stress in some regions.

The best protection against climate-related conflict is the same as the best protection against conflict generally: inclusive governance, fair access to resources, effective conflict resolution, and economic opportunities. Climate adaptation programmes that strengthen these foundations — for example through participatory water management or equitable land rights — contribute to peace as well as resilience.

Infrastructure: the hidden backbone

Modern life depends on networks: electricity grids, water and sewerage systems, roads, bridges, ports, telecommunications and the internet. Much of this infrastructure was designed for the climate of the past. Drains are sized for historical rainfall; bridges for historical flood levels; power lines and transformers for historical temperatures.

Climate change pushes these systems beyond their design limits. Heat reduces the efficiency of power lines and transformers just when demand for cooling peaks. Floods inundate substations, pumping stations and roads. Landslides cut transport and communication links. Sea-level rise threatens ports, coastal roads and airports. Because infrastructure systems are interconnected, the failure of one can trigger others: a power cut disables water pumps and mobile phone towers; a flooded road prevents repair crews from reaching a damaged substation.

Building resilient infrastructure means designing for future climate conditions rather than past ones, adding redundancy (more than one route, more than one source), protecting critical facilities (raising substations and generators above flood levels), maintaining existing assets, and planning for rapid recovery. It also means using nature: mangroves, wetlands and forests protect infrastructure at low cost.

Systemic risk and the role of finance

Financial regulators increasingly recognise climate change as a source of systemic risk. Physical risks — damage to property and assets — can threaten banks and insurers. Transition risks arise as the world shifts away from fossil fuels, potentially leaving coal plants, mines and related assets "stranded". Indonesia's financial regulator (OJK) has developed a sustainable finance taxonomy and is introducing climate risk management requirements for banks. For households, the same logic applies at a smaller scale: assets in high-risk locations may lose value, and diversification protects against shocks (Chapter 12).

Breaking the cascade

Cascading risks can be interrupted. The key principles are:

- **Redundancy** — backup systems, alternative routes, multiple suppliers.
- **Diversity** — diverse crops, energy sources and income streams.
- **Modularity** — systems that can fail in parts without collapsing entirely, such as local solar mini-grids.
- **Early warning** — detecting problems early enough to act.
- **Social protection** — cash transfers and safety nets that prevent a shock from forcing families into irreversible losses, such as selling productive assets or withdrawing children from school. Indonesia's

social protection programmes, including *Program Keluarga Harapan* and food assistance, can be scaled up rapidly after disasters if designed to be "adaptive".

- **Coordination** — breaking down silos between agencies responsible for different sectors.

KEY POINTS

- Climate risks are compound and cascading; impacts in one sector spread to others.
- Prevention pays: early warning and resilient infrastructure deliver large returns.
- Climate change could drive up to 216 million internal migrants by 2050; well-supported migration can be adaptive.
- Climate acts as a threat multiplier for conflict, especially where institutions are weak.
- Infrastructure designed for past climates must be upgraded for future ones.

PART III

Preparing Households

CHAPTER 9

Know Your Risk: A Personal and Local Assessment

Preparation begins with knowing what you are preparing for. A family on a riverbank faces different threats from a family on a hillside or in a high-rise apartment. An hour spent assessing your risks will guide every decision that follows.

Risk is more than hazard

Disaster specialists describe risk as the product of three factors:

- **Hazard** — the event that could cause harm: a flood, a heat wave, a landslide, a drought.
- **Exposure** — whether people and assets are in harm's way: living in the floodplain, working outdoors, owning a house on a slope.
- **Vulnerability** — how susceptible those people and assets are to harm: the strength of the house, the health of the people, their savings, knowledge and social connections.

This framework is empowering because it shows that risk can be reduced even when the hazard cannot be prevented. We cannot stop the monsoon, but we can avoid building in the floodplain (reducing exposure) or raise the floor and keep documents in waterproof bags (reducing vulnerability). We can also strengthen **capacity**: the skills, resources and networks that allow people to cope and recover.

Step 1: Identify your hazards

Start by listing the hazards that could affect your home, workplace and the places your family spends time, such as schools. Consider both cli-

mate-related and other hazards — in Indonesia, earthquakes, tsunamis and volcanic eruptions must be part of any household plan.

Useful sources of information include:

- **InaRISK** (inarisk.bnpb.go.id and the InaRISK Personal app) — BNPB's portal showing hazard and risk maps for floods, landslides, earthquakes, tsunamis, volcanic eruptions, droughts, forest fires, extreme weather and more, down to the village level.
- **BMKG** (bmkg.go.id and the *Info BMKG* app) — weather forecasts, early warnings for extreme weather, seasonal climate outlooks, earthquake and tsunami information, and air quality.
- **MAGMA Indonesia** (magma.esdm.go.id) — volcanic activity and alert levels from the Centre for Volcanology and Geological Hazard Mitigation (PVMBG), as well as landslide susceptibility information.
- **Local BPBD** (regional disaster management agency) — local hazard maps, evacuation routes and contingency plans.
- **Your own community** — older residents remember past floods, landslides and droughts: how high the water reached, which roads were cut, which slopes moved. This knowledge is invaluable.

Step 2: Assess your exposure

For each hazard, ask how exposed your household is.

- **Flood:** Is your home in a low-lying area, near a river, canal or coast? Has the neighbourhood flooded before? How high did the water reach? Is the ground floor below street level?
- **Heat:** Is your home poorly ventilated, under a metal roof, or on a top floor? Do family members work outdoors? Is there shade and green space nearby?
- **Landslide:** Is your home on or below a steep slope? Are there cracks in the ground or walls? Has the slope been cut for construction?
- **Drought and water:** Does your water come from a shallow well that dries up in the dry season? Is piped water reliable?
- **Fire and haze:** Do you live near peatlands, forests or plantations that burn in dry years?
- **Coastal hazards:** Are you within reach of tidal flooding, storm surges or tsunami inundation zones?

Step 3: Assess your vulnerability and capacity

Next, consider the characteristics of your household that make it more or less able to cope:

- **People:** Are there infants, older people, pregnant women, people with disabilities or chronic illnesses, or people who need regular medication? Who would need help to evacuate?
- **Building:** What is the house made of? Is it built to withstand earthquakes and strong winds? Are electrical outlets and appliances above likely flood levels?
- **Finances:** Do you have savings to cover an emergency? Is your income dependent on weather — farming, fishing, outdoor trade? Do you have insurance?
- **Knowledge and skills:** Does everyone know the warning signs, evacuation routes and basic first aid? Can someone swim?
- **Connections:** Do you know your neighbours? Is there a community organisation, RT/RW leadership, mosque, church or village disaster team that would help?

Step 4: Rank and prioritise

With this information, rank your risks. A simple method is to score each hazard on likelihood (how often might it happen?) and impact (how bad would it be?), each from 1 to 5, and multiply the two. Focus first on the highest scores, but do not ignore low-likelihood, high-impact events such as tsunamis in coastal areas — these require planning even if they are rare.

Hazard	Likelihood (1–5)	Impact (1–5)	Score	Priority actions
Urban flood	4	3	12	Raise valuables; flood plan; waterproof documents
Heat wave	5	2	10	Improve ventilation; check on grandmother
Dengue	4	3	12	Weekly 3M Plus; screens; know warning signs
Earthquake	2	5	10	Secure furniture; family meeting point

An example household risk matrix. Your own scores will differ.

Step 5: Look ahead

Climate change means that past experience is an imperfect guide. A neighbourhood that flooded once in twenty years may flood every few years in the future; a well that has never run dry may fail in a severe El Niño. When making long-term decisions — buying land, building or renovating a house, choosing a school — consider how risks may change over the next 20–30 years. Ask whether the site is likely to be affected by sea-level rise, more intense rainfall or subsidence. If in doubt, choose the safer option.

INDONESIA FOCUS: WARNING SIGNS FROM NATURE AND COMMUNITY

Many Indonesian communities hold detailed knowledge of local warning signs: the smell or colour of river water before a flash flood, the behaviour of animals before an eruption, the withdrawal of the sea before a tsunami, cracks in rice-field terraces before a landslide. The story of *smong* on Simeulue Island (Chapter 13) is the most famous example. Combining this knowledge with modern warnings from BMKG and BPBD gives communities two independent sources of alerts — redundancy that can save lives.

Revisit regularly

A risk assessment is not a one-time exercise. Revisit it at least once a year — ideally before the rainy season — and whenever your circumstances change: a new baby, an older parent moving in, a new house or job, or new information about local hazards. Keep a written copy with your emergency plan.

TAKE ACTION: YOUR ONE-HOUR RISK ASSESSMENT

1. Look up your home address on InaRISK and note the hazard levels.
2. Install the *Info BMKG* app and enable notifications for your area.
3. Ask an older neighbour about past disasters in your area.
4. List the people in your household who would need extra help.
5. Fill in a simple risk matrix like the one above.
6. Choose the top three actions you will take in the next month.

KEY POINTS

- Risk = hazard × exposure × vulnerability; you can reduce risk by reducing exposure and vulnerability.

- Use InaRISK, BMKG, MAGMA and your local BPBD, plus community memory.
- Assess people, buildings, finances, skills and connections.
- Rank risks by likelihood and impact; plan for rare but catastrophic events too.
- Climate change means the future may differ from the past; revisit your assessment yearly.

CHAPTER 10

The Resilient Household: Water, Food, Energy and Shelter

A resilient household can keep its members safe, fed, hydrated and reasonably comfortable for several days without outside help — and can recover quickly afterwards. Achieving this is less about expensive equipment than about planning and habit.

The three-day minimum, the two-week goal

After a major disaster, it can take days for help to arrive. Roads may be cut, electricity and water supplies interrupted, shops closed and emergency services overwhelmed. Disaster agencies around the world recommend that households be able to sustain themselves for at least three days. In areas exposed to major hazards — earthquakes, large floods, volcanic eruptions — aiming for one to two weeks of supplies is sensible.

This does not require a bunker full of supplies. It means keeping a modest, rotated stock of essentials and knowing how to use them.

Water

Water is the most critical supply. A person can survive weeks without food but only a few days without water, and less in hot conditions.

How much? Plan for at least 4 litres per person per day — about 2–3 litres for drinking and the rest for cooking and minimal hygiene. In hot weather, for nursing mothers, sick people and those doing physical work, more is needed. The Sphere humanitarian standards set 15 litres per person per day as the minimum for all basic needs in emergency

settings (Sphere Association, 2018). For a family of four, a three-day minimum supply is therefore around 50 litres for drinking and cooking.

Storage. Use clean, food-grade containers with tight lids, stored in a cool, dark place. Commercially bottled water or sealed gallons (*galon*) are convenient. Rotate stored water every six months. Covered tanks for rainwater harvesting provide a larger backup for washing and flushing. Always cover containers to prevent mosquito breeding.

Treatment. When the safety of water is uncertain:

- **Boiling** — bring water to a rolling boil for at least one minute (longer at high altitude). This is the most reliable method.
- **Chlorination** — use water purification tablets or drops according to the instructions, or unscented household bleach in the dose recommended by health authorities, and wait at least 30 minutes.
- **Filtration** — ceramic and membrane filters remove many pathogens; use them in combination with chlorination for best results.
- **Solar disinfection (SODIS)** — fill clear plastic (PET) bottles with clear water and leave them in full sunlight for at least six hours (two days if cloudy).

Let cloudy water settle and pour it through a clean cloth before treatment.

Food

Keep a stock of non-perishable food that your family will actually eat and that requires little or no cooking or water. Good options in an Indonesian kitchen include:

- rice, instant noodles (in moderation), oats and biscuits;
- canned fish (sardines, tuna), canned meat, canned beans and corned beef;
- *abon* (shredded dried meat or fish), *kering tempe*, peanuts and dried fruit;
- powdered or UHT milk, especially for children;
- sugar, salt, cooking oil, instant seasoning, coffee and tea;
- special foods for infants, older people or those with dietary needs.

Follow the “first in, first out” principle: use the oldest items in daily cooking and replace them, so the stock stays fresh. Keep a manual can opener. Plan for a way to cook without electricity or piped gas: a small LPG stove with a spare canister, or a charcoal stove used only outdoors, never indoors because of carbon monoxide.

Energy and light

Power cuts accompany most disasters, and extreme heat increasingly strains electricity grids. Prepare with:

- **Light** — LED flashlights and headlamps with spare batteries; solar or rechargeable lanterns. Avoid candles where possible because of fire risk.
- **Communication** — a power bank (kept charged), a battery-powered or hand-crank radio for official announcements, and a list of important phone numbers on paper.
- **Small-scale solar** — a solar panel kit with a battery can power lights, phones and a fan. Costs have fallen dramatically, and such systems are useful in daily life as well as emergencies.
- **Generators** — if you use one, operate it outdoors, far from windows and doors, because exhaust contains deadly carbon monoxide.

Shelter: making the home safer

Your home is your first line of defence. Improvements need not be expensive.

Against heat:

- Improve ventilation with openings on opposite walls for cross-breezes; vents near the roof let hot air escape.
- Shade windows and walls with awnings, blinds, trees or climbing plants.
- Paint roofs and walls in light colours, or install a reflective layer under metal roofs. Ceiling insulation greatly reduces heat from metal or concrete roofs.
- Use ceiling or standing fans; they consume far less electricity than air conditioners and are effective up to quite high temperatures in humid conditions, provided the person can sweat and drink.
- If you have air conditioning, set it to a moderate temperature (around 25–27 °C) to reduce costs and strain on the grid.

Against flooding:

- Raise electrical outlets, meters and key appliances above likely flood levels.
- Install one-way valves on drains to prevent sewage backflow.
- Keep a place on an upper floor or high shelf for documents, valuables and emergency supplies.

- Use water-resistant materials (tiles, concrete, raised wooden furniture) on ground floors in flood-prone homes.
- Keep gutters and drains clear; create infiltration wells or biopores in the yard.
- For new houses or major renovations in flood-prone areas, consider raising the floor level or building on stilts, as traditional houses in many coastal and river communities do.

Against wind, earthquakes and landslides:

- Make sure roof sheets and tiles are securely fixed.
- Secure tall furniture, water heaters and gas cylinders to walls.
- Follow earthquake-resistant building practices: reinforced concrete frames with proper ties between columns, beams and foundations. The Ministry of Public Works publishes guidelines for safe houses (*rumah tahan gempa*).
- Do not cut steep slopes behind or below the house without engineering advice; plant deep-rooted vegetation; manage drainage so water does not pool on slopes.

Sanitation and hygiene

When water and sewerage systems fail, disease spreads quickly. Keep soap, hand sanitiser, wet wipes, sanitary pads, diapers if needed, strong plastic bags and toilet paper in your supplies. Know how to create an emergency toilet: a bucket lined with a heavy plastic bag, with a little lime, ash or soil added after each use, and sealed bags disposed of safely.

Medicines and first aid

Keep a well-stocked first-aid kit (see Chapter 11) and at least a two-week supply of any regular prescription medicines, along with copies of prescriptions. Medicines can be damaged by heat; store them in the coolest part of the house.

Keep it practical

The best preparations are those that serve daily life as well as emergencies. Rainwater tanks reduce water bills; solar lights reduce electricity bills; shade trees cool the house every day; a pantry with a rotated stock smooths out food price fluctuations. When preparation becomes part of normal household routine, it is far more likely to be maintained.

TAKE ACTION: HOUSEHOLD RESILIENCE IN FIVE STEPS

1. Store at least 4 litres of drinking water per person per day for three days, and rotate it.
2. Build a three-day (then two-week) food stock using "first in, first out".
3. Buy a flashlight, power bank and battery radio; consider a small solar kit.
4. Choose one heat-proofing and one flood-proofing improvement for your home this year.
5. Put regular medicines, documents and supplies in a place that stays dry.

KEY POINTS

- Aim for at least three days of self-sufficiency; one to two weeks in high-risk areas.
- Water first: 4 litres per person per day minimum; know how to treat water.
- Rotate non-perishable food; have a safe way to cook without electricity.
- Simple home improvements reduce heat and flood damage.
- Preparation works best when it serves everyday life too.

CHAPTER 11

Emergencies: Plans, Kits, Evacuation and First Aid

In an emergency, there is no time to think from scratch. People fall back on what they have rehearsed. A family plan, a packed bag and a few first-aid skills turn panic into action.

The family emergency plan

A family emergency plan answers a few simple questions in advance, so that everyone knows what to do even if they are separated or communication networks are down.

1. **How will we receive warnings?** Identify official channels — the *Info BMKG* app, local BPBD social media, radio, mosque loudspeakers, village sirens or *kentongan* — and make sure at least one adult receives alerts on their phone.
2. **Where will we go?** Identify a safe place inside the home (for example an upper floor during floods), a meeting point near home (a field, a mosque), and an evacuation destination outside the neighbourhood (a relative's house, a designated evacuation site) for each major hazard. Tsunami evacuation requires reaching high ground or a designated vertical evacuation building quickly — know the route.
3. **How will we get there?** Plan at least two routes. Know whether you will walk, ride or drive, and what to do if roads are blocked.
4. **How will we communicate?** Choose an out-of-area contact — a relative in another city — whom everyone will call or message to report their status, since local networks are often overloaded. Write

important numbers on paper. Agree to use text messages, which get through more easily than calls when networks are congested.

5. **Who needs help?** Assign responsibilities: who picks up the children from school, who helps grandparents or a family member with a disability, who takes the pets, who grabs the emergency bag.
6. **What about school and work?** Learn the emergency plans of your children's schools and your workplace, including how children will be released to parents.

Write the plan down, keep copies in the emergency bag and on everyone's phone, and discuss it with all members of the household, including children, in a calm and practical way.

Practise

A plan that has never been practised is likely to fail. Hold a short drill at least once or twice a year — before the rainy season is a good moment. Walk the evacuation route, time how long it takes, check the emergency bag, and test the communication plan. Schools and communities in Indonesia increasingly hold earthquake and tsunami drills; participate when they do. Children who have practised are far less likely to panic.

The emergency bag (go-bag)

An emergency bag — sometimes called a go-bag or *tas siaga bencana* — is a pre-packed bag that allows you to leave home within minutes and survive for about three days. Keep one for each family member (or one larger bag per household) in an accessible place near the exit.

EMERGENCY BAG CHECKLIST

- Drinking water (at least 1–2 litres per person) and high-energy food that needs no cooking
- Copies of important documents in a waterproof bag: ID cards (KTP), family card (*Kartu Keluarga*), birth certificates, land and house certificates, insurance policies, BPJS cards, bank details, prescriptions — plus digital copies in secure cloud storage
- Some cash in small denominations
- Flashlight or headlamp with spare batteries; power bank and charging cables
- Battery or hand-crank radio; whistle to signal for help
- First-aid kit and a supply of regular medicines
- Change of clothes, raincoat, sandals or sturdy shoes, blanket or sarong

- Masks (N95 for haze or ash), hand sanitiser, soap, toothbrush, sanitary pads
- Items for infants, older people or people with disabilities: formula, diapers, spare glasses, hearing-aid batteries, mobility aids
- Paper list of emergency contacts and a copy of the family plan
- Multi-purpose knife, duct tape, plastic sheet, strong bags

Check the bag every six months: replace expired food, water, medicines and batteries, and update documents and clothes as children grow.

Evacuation

When authorities issue an evacuation order, follow it. Many disaster deaths occur among people who stayed too long. Evacuation is easier and safer when started early, before roads flood, darkness falls or panic sets in.

Before leaving: if there is time, switch off electricity, gas and water at the main supply; lock the house; take the emergency bag; and inform your out-of-area contact.

On the way: follow designated routes; avoid floodwater, fallen power lines and unstable slopes; help neighbours who need assistance.

At the evacuation site: register with the officials so relatives can find you; follow hygiene rules; look after children; and report health problems early. Evacuation shelters can be crowded and stressful; women, children, older people and people with disabilities have particular needs for safety, privacy and accessibility, and community organisers should plan for these.

Tsunami: in coastal areas, a strong or long earthquake (one that makes it hard to stand, or lasts more than about 20 seconds), an unusual roar from the sea, or a sudden withdrawal of the sea are natural warnings. Do not wait for an official alert: move immediately to high ground or inland, on foot if possible, and stay there until authorities declare it safe. Tsunamis come in a series of waves, and the first is often not the largest.

Basic first aid

Professional medical help may be delayed after a disaster. Basic first aid skills can save lives in the critical minutes and hours. Organisations

such as the Indonesian Red Cross (PMI) offer first-aid training; it is one of the best investments a family can make.

Key skills include:

- **Checking safety and calling for help** — make sure the scene is safe before approaching; call 112 or 119.
- **CPR (cardiopulmonary resuscitation)** — for a person who is unresponsive and not breathing normally: push hard and fast in the centre of the chest, about 100–120 compressions per minute, and use an automated external defibrillator (AED) if available.
- **Controlling bleeding** — apply firm, direct pressure to the wound with a clean cloth; keep pressing.
- **Treating burns** — cool with clean running water for at least 20 minutes; do not apply toothpaste, butter or ice.
- **Recognising and treating heat illness** — see Chapter 4.
- **Treating dehydration** — oral rehydration solution.
- **Stabilising fractures** — keep the injured limb still with a splint or sling.
- **Recovery position** — for an unconscious person who is breathing, to keep the airway open.

FIRST-AID KIT CONTENTS

Sterile gauze and bandages of various sizes; adhesive plasters; elastic bandage; triangular bandage; medical tape; scissors and tweezers; antiseptic solution or wipes; disposable gloves; thermometer; oral rehydration salts (*oralit*); paracetamol; antihistamine; burn gel; insect repellent; a first-aid manual; and any personal medicines.

After the emergency

Recovery begins as soon as the immediate danger passes. Return home only when authorities say it is safe. Inspect the house for structural damage, gas leaks and electrical hazards before entering; if in doubt, ask for professional help. Photograph damage for insurance and assistance claims. Clean and disinfect thoroughly after floods. Watch for signs of illness in the days and weeks after — fever, diarrhoea, skin infections — and for signs of psychological distress in family members, especially children (see Chapter 16). Finally, review what worked and what did not, and update your plan.

KEY POINTS

- Make a written family plan: warnings, safe places, routes, communication, responsibilities.
- Practise at least once or twice a year.
- Keep a packed emergency bag with water, food, documents, cash, light, radio and first aid.
- Evacuate early when instructed; for tsunamis, natural warning signs mean go now.
- Learn first aid and CPR; keep a stocked first-aid kit.

CHAPTER 12

Financial Resilience

Disasters destroy property, but they also destroy finances — and financial damage often lasts far longer than physical damage. Money that is saved, insured and diversified before a shock is the fastest route to recovery afterwards.

Why financial preparation matters

After a disaster, households face a double blow: sudden expenses (repairs, medical bills, temporary accommodation, replacing belongings) and lost income (damaged businesses, destroyed crops, days off work). Families without savings or insurance are forced into harmful coping strategies: borrowing at high interest rates from informal lenders, selling productive assets such as livestock, tools or land, cutting food consumption, or taking children out of school. These choices can trap families in poverty long after the floodwaters recede.

Research on disasters consistently finds that the poorest households lose a smaller absolute amount but a much larger share of their wealth, and take far longer to recover. Financial resilience is therefore not a luxury; it is a central part of climate adaptation.

Build an emergency fund

The foundation of financial resilience is an emergency fund: money set aside specifically for unexpected events, kept in a safe and accessible form. Financial planners commonly recommend saving enough to cover three to six months of essential expenses; for households with irregular incomes, such as farmers, fishers and informal traders, a larger buffer is wise.

For many families, that target seems out of reach. Start small. Setting aside a fixed amount each week or month — even a few tens of thousands of rupiah — builds the habit, and the fund grows over time. Practical tips:

- Keep emergency savings separate from daily spending money, in a savings account that is accessible but not linked to everyday spending.
- Keep a small amount of cash at home in your emergency bag, because ATMs and digital payments may not work during power or network outages.
- Use community savings groups (*arisan*, cooperatives, credit unions) where trusted structures exist.
- Do not use the emergency fund for non-emergencies; replenish it after use.

Protect your documents and digital access

Lost documents can block access to bank accounts, insurance payments, government aid, land rights and medical care. Keep originals in a waterproof, fire-resistant container in a high place, copies in the emergency bag, and scanned copies in secure cloud storage or an encrypted drive. Important documents include identity cards (KTP), family cards (KK), birth and marriage certificates, land and house certificates (*sertifikat tanah*), vehicle documents (BPKB, STNK), insurance policies, BPJS cards, bank account details, educational certificates and medical records.

Write down important account numbers and contact details for your bank and insurer. Enable secure access to mobile banking, and make sure at least one trusted family member knows how to access essential accounts in an emergency.

Insurance

Insurance transfers the financial risk of rare but costly events to an insurer in exchange for regular premiums. It is most valuable for losses that would be devastating if they occurred: the destruction of a house, a serious illness, the death of a breadwinner.

Health insurance. BPJS Kesehatan, Indonesia's national health insurance programme, covers a wide range of medical care. Make sure every family member is registered and that contributions are up to date.

Property insurance. Standard property insurance policies in Indonesia typically cover fire and some other risks, but flood, earthquake and other natural hazards often require additional coverage. Read the policy carefully to see what is and is not covered.

Agricultural insurance. Farmers can access subsidised rice farm insurance (*Asuransi Usaha Tani Padi*, AUTP) and livestock insurance, which compensate for losses from floods, droughts, pests and disease. Ask the local agriculture office or farmers' group about eligibility.

Life insurance. For families dependent on one or two earners, life insurance provides protection if a breadwinner dies.

Microinsurance and parametric insurance. Newer products pay out automatically when a measured trigger — such as rainfall below a threshold or wind speed above a level — is reached, without the need for damage assessment. They can provide fast, low-cost protection, though they may not match actual losses exactly.

At the national level, Indonesia has developed a disaster risk financing strategy, including insurance for state assets and a pooled disaster fund (*Dana Bersama Penanggulangan Bencana*) established by Presidential Regulation No. 75 of 2021, to ensure that public funds are available quickly after disasters.

Diversify income and assets

Households that depend on a single source of income — especially one tied to the weather — are highly vulnerable. Diversification spreads risk:

- Combine farm and off-farm income where possible.
- Develop skills that are useful in multiple settings.
- Hold savings in more than one form and place.
- Avoid concentrating all wealth in a single asset in a high-risk location, such as a house in a flood-prone area.

Manage debt wisely

Debt can help households recover, but expensive debt can trap them. Avoid high-interest informal lenders and illegal online lending (*pinjol ilegal*), which often prey on people in distress; check whether lenders are registered with OJK, Indonesia's financial services authority. After disasters, banks and microfinance institutions sometimes offer loan restructuring or payment holidays; ask before falling behind on payments.

Know what help is available

After major disasters, governments, humanitarian organisations and community groups provide assistance: emergency relief, cash transfers, housing reconstruction grants and livelihood support. Registering with authorities at evacuation sites, keeping documents accessible and documenting damage with photographs make it easier to receive help. Social protection programmes such as *Program Keluarga Harapan* (PKH) and food assistance programmes may be expanded after disasters.

Think long term

Climate change affects long-term financial decisions as well as emergency preparedness. Before buying land or a house, check hazard maps and ask about past floods and landslides. Consider how property values in high-risk areas may change as risks become better understood and insurance becomes more expensive. When choosing careers or businesses, consider how they may be affected by climate change — and the opportunities that the transition to a low-carbon economy creates in fields such as renewable energy, efficient buildings, sustainable agriculture and climate services.

TAKE ACTION: YOUR FINANCIAL RESILIENCE CHECKLIST

- Open a separate emergency savings account and set up a regular deposit, however small.
- Keep some cash in your emergency bag.
- Scan all important documents and store them securely in the cloud.
- Check that every family member is registered with BPJS Kesehatan.
- Review your property insurance: does it cover flood and earthquake?
- If you farm, ask about AUTP crop insurance.
- Check any lender's registration with OJK before borrowing.

KEY POINTS

- Financial shocks after disasters can trap families in poverty; preparation prevents this.
- Build an emergency fund, starting small; keep some cash at home.
- Protect documents physically and digitally.
- Use insurance for catastrophic risks: health, property, crops and life.
- Diversify income and avoid high-interest debt.

PART IV

Stronger Together

CHAPTER 13

Community Resilience: Social Capital, Mutual Aid and Local Knowledge

In almost every disaster, the first responders are not professionals but neighbours. The strength of the ties between people — who knows whom, who trusts whom, who will knock on whose door — is one of the most powerful predictors of who survives and how quickly a place recovers.

The hidden infrastructure

When researchers study why some communities recover from disasters faster than others, they often find that physical damage and financial aid explain less than expected. A major factor is social capital: the networks, norms and trust that enable people to act together. The political scientist Daniel Aldrich, studying recovery after disasters in Japan, India and the United States, found that communities with dense social networks recovered faster and more fully than those with weaker ties, even when they received similar outside assistance (Aldrich, 2012).

Social capital takes different forms:

- **Bonding** ties connect people who are similar — family, close friends, neighbours. They provide immediate support in a crisis: shelter, food, childcare, emotional comfort.
- **Bridging** ties connect people across different groups — ethnic, religious, economic. They bring in new information and resources and help communities avoid divisions.

- **Linking** ties connect communities to institutions with power and resources — local government, agencies, NGOs, businesses. They help communities access aid and influence decisions.

Resilient communities have all three.

Gotong royong: an Indonesian strength

Indonesia has a deep tradition of collective action, expressed in the concept of *gotong royong* — mutual cooperation, working together to lighten a shared burden. It appears in communal work to clean drains and village roads (*kerja bakti*), in the rotating savings groups of *arisan*, in the neighbourhood structures of RT and RW, in religious communities and in countless informal networks of support.

After disasters in Indonesia, these traditions have repeatedly proved their value: neighbours sheltering neighbours, communal kitchens (*dapur umum*) feeding the displaced, volunteers clearing debris, and religious and community organisations mobilising relief faster than formal systems. The challenge is to sustain and strengthen these networks before disasters strike, and to link them with formal early warning and response systems.

The lesson of smong

On 26 December 2004, a magnitude 9.1 earthquake off the west coast of Sumatra triggered a tsunami that killed more than 160,000 people in Indonesia alone, most in Aceh. Simeulue Island lay only about 60 kilometres from the epicentre, and waves exceeding 10 metres struck parts of its coast within minutes. Yet only seven of the island's roughly 78,000 inhabitants died in the tsunami (McAdoo et al., 2006).

The reason was *smong* — the local word for tsunami in the Devayan language, and the name of a body of oral tradition passed down since a deadly tsunami in 1907. Through lullabies, poems and stories told by grandparents, the people of Simeulue had learned that when a strong earthquake strikes and the sea withdraws, one must run immediately to the hills. When the ground shook in 2004, the cry of "*smong!*" spread along the coast, and people fled to higher ground.

The *smong* story shows that local knowledge, embedded in culture and repeated across generations, can be as effective as any technology. It also shows the importance of keeping such knowledge alive and connecting it to modern science.

INDONESIA FOCUS: DISASTER RESILIENT VILLAGES

BNPB's *Desa/Kelurahan Tangguh Bencana* (Destana) programme, based on BNPB Regulation No. 1 of 2012, supports villages in assessing their risks, preparing contingency plans and evacuation routes, forming village disaster preparedness teams, establishing early warning systems and integrating disaster risk reduction into village development plans and budgets. The *Program Kampung Iklim* (ProKlim), run by the Ministry of Environment, recognises and supports communities that undertake climate adaptation and mitigation actions such as water conservation, waste management, urban farming and tree planting. Both programmes provide a framework for organising community action.

Mutual aid

Mutual aid is the practice of people organising to meet each other's needs directly, on the basis of solidarity rather than charity. It ranges from informal arrangements — neighbours sharing food, checking on older residents, organising childcare — to structured networks with shared resources and plans.

Mutual aid became widely visible during the COVID-19 pandemic, when communities around the world, including in Indonesia, organised food distribution, support for people in isolation and information sharing. The same networks can be mobilised for climate shocks.

Effective mutual aid networks typically:

- map the needs and skills in the community — who has a boat, a generator, medical training, a vehicle; who lives alone, has a disability or needs regular medicine;
- establish clear communication channels, such as neighbourhood messaging groups and phone trees;
- maintain shared resources — tools, first-aid supplies, a communal kitchen, a store of food;
- meet regularly, not just in crises, to build trust;
- connect with local government and organisations to avoid duplication and access support.

Inclusion: leaving no one behind

Disasters do not affect everyone equally. Women, children, older people, people with disabilities, the poor, migrants and marginalised groups often face greater risks and receive less help. In many disasters,

women die at higher rates than men because of differences in mobility, access to warnings, clothing, caregiving responsibilities and the ability to swim. People with disabilities may be unable to hear sirens, see warning signs or reach evacuation sites without help.

Inclusive preparedness means involving these groups in planning, not merely planning for them. It means accessible warnings (visual, audible, in local languages), accessible evacuation routes and shelters, attention to safety and privacy for women and girls in shelters, and registration systems that identify who will need assistance.

Local knowledge and science together

Indigenous and local knowledge offers detailed, place-based understanding of environmental change accumulated over generations. Examples across Indonesia include traditional calendars that guide planting and fishing, such as the *pranata mangsa* of Javanese farmers; customary rules (*sasi*) in Maluku and Papua that close fishing grounds or harvests for periods to allow recovery; the *subak* irrigation system of Bali, a UNESCO World Heritage cultural landscape that manages water cooperatively through water temples; and traditional house designs adapted to earthquakes, floods and heat.

Such knowledge is not infallible, and some traditional indicators may become less reliable as the climate shifts beyond historical experience. The most robust approach combines local knowledge with scientific data — for example, climate field schools in which farmers learn to interpret seasonal forecasts alongside their own observations. Respecting and documenting local knowledge, and ensuring that communities have a voice in adaptation decisions, are both ethical obligations and practical necessities.

Building community resilience: where to start

Community resilience does not require waiting for a government programme. Any group of neighbours can begin:

1. **Get together.** Use an existing gathering — an RT meeting, a religious study group, a *kerja bakti* — to discuss local risks.
2. **Map risks and resources.** Draw a simple map of the neighbourhood showing hazards (flood-prone streets, unstable slopes), safe places, evacuation routes, and resources (people with skills, equipment, vehicles).

3. **Identify who needs help.** Make a confidential list of residents who will need assistance in an emergency, and assign buddies.
4. **Set up communication.** Create a messaging group and a phone tree; agree on warning signals.
5. **Act on prevention.** Clean drains, plant trees, eliminate mosquito breeding sites, build infiltration wells.
6. **Practise.** Hold a simple evacuation drill.
7. **Connect.** Contact the local BPBD, village government and health centre (*puskesmas*) to link your efforts with official systems.

KEY POINTS

- Social capital — bonding, bridging and linking ties — strongly predicts disaster recovery.
- *Gotong royong* and community structures are an Indonesian strength to cultivate.
- *Smong* on Simeulue shows how local knowledge saves lives.
- Mutual aid networks map needs and resources and act together.
- Inclusion of women, older people, people with disabilities and the marginalised is essential.

CHAPTER 14

Cities and Villages That Adapt

More than half of Indonesians now live in cities, and the share keeps growing. Yet the country's food, water and much of its resilience still depend on villages, forests and coasts. Adaptation must work in both.

The urban challenge

Cities concentrate people, wealth and infrastructure — and therefore risk. They also concentrate the capacity to respond: institutions, expertise, finance and innovation. Indonesian cities face a distinctive mix of climate hazards: urban heat islands, flash and pluvial flooding, tidal flooding and subsidence on the coast, water shortages, air pollution and disease. These hazards hit hardest in informal settlements (*kampung*) along rivers, railways and coasts, where housing is crowded, drainage poor and services limited.

Principles of urban resilience

Plan with risk in mind. Spatial plans (*Rencana Tata Ruang Wilayah*) should keep new development out of floodplains, riverbanks, coastal hazard zones and unstable slopes, and protect green spaces and water catchments. Enforcement is as important as planning.

Work with water, not only against it. Instead of relying solely on walls and pumps, cities can create space for water: restored rivers with natural banks, retention ponds, wetlands, green corridors and permeable surfaces. These measures reduce flooding, recharge groundwater, cool the city and provide recreational space.

Stop the sinking. In coastal cities affected by subsidence, the single most important measure is to replace groundwater extraction with reliable piped water supply from surface sources, combined with regu-

lation and enforcement. Sea walls can buy time but do not address the underlying cause, and they can fail catastrophically.

Cool the city. Urban greening — street trees, parks, green roofs and walls — reduces temperatures, especially at night. Cool roofs and pavements reflect sunlight. Building codes can promote ventilation and shading. Shaded walkways, markets and public transport stops protect people who spend time outdoors.

Upgrade informal settlements. Programmes that improve drainage, water supply, sanitation, paths and housing in existing *kampung* — with residents' participation — reduce risk more effectively and humanely than evictions. Indonesia's long history of kampung improvement, beginning with the Kampung Improvement Program in Jakarta and Surabaya in the 1970s, offers lessons. Surabaya, in particular, has been recognised for community-based greening and cleanliness campaigns.

Strengthen services. Reliable water, electricity, health care, waste collection and public transport are themselves forms of resilience. A city where waste is collected has fewer clogged drains and mosquito breeding sites; a city with good public transport is less vulnerable to fuel shortages and emits less.

Plan for heat and health. Heat action plans, cooling centres and surveillance of heat-related illness belong in every city's disaster plan.

Nature-based solutions on the coast

Mangrove forests are among the most valuable ecosystems for climate resilience. Their dense roots reduce wave energy and erosion, trap sediment that can help coastlines keep pace with rising seas, provide nursery habitats for fish and shrimp, and store large amounts of carbon in their soils. Studies show that mangrove belts can significantly reduce wave heights and storm surge damage behind them (Spalding et al., 2014).

Indonesia has the world's largest area of mangroves — around 3.4 million hectares, roughly a fifth of the global total — but large areas have been lost to aquaculture ponds, agriculture and coastal development. National programmes now aim to rehabilitate hundreds of thousands of hectares. Success depends on planting the right species in the right places (where tidal conditions suit mangroves), involving local communities, and addressing the causes of loss. Poorly planned planting — for example on mudflats where mangroves never grew — often fails.

Other coastal approaches include "hybrid" engineering that combines natural and built elements, such as permeable bamboo structures that trap sediment and allow mangroves to re-establish, as tested along Central Java's coast. Seagrass beds and coral reefs also protect coasts from waves; their health depends on reducing pollution, destructive fishing and, ultimately, warming.

Rural resilience

Rural communities depend directly on natural resources and the weather. Their resilience rests on several foundations:

Water. Small reservoirs (*embung*), check dams, rainwater harvesting, spring protection and catchment reforestation secure water for crops, livestock and households through dry seasons.

Diversified farming. Agroforestry, crop diversity, integrated rice-fish systems and home gardens (*pekarangan*) reduce dependence on a single crop and spread risk.

Soil and land. Terracing, contour planting, cover crops and organic matter reduce erosion and landslides, retain moisture and maintain fertility. Protecting forests on steep slopes is critical for preventing landslides and flash floods downstream.

Information. Access to weather forecasts, early warnings, market prices and agricultural advice through mobile phones, extension officers and farmers' groups helps rural households make better decisions.

Finance and markets. Savings groups, cooperatives, crop insurance and access to markets help farmers manage risk and invest in adaptation.

Village governance. Since the Village Law of 2014, Indonesian villages receive direct transfers (*Dana Desa*) and have considerable autonomy over their use. Villages can use these funds for adaptation: drainage, water infrastructure, early warning, mangrove and forest restoration, and training. Integrating risk assessments into village development plans (*RPJMDes*) ensures that funds are directed where they reduce risk most.

Peatlands and forests

Indonesia's forests and peatlands are both a global climate asset and a local resilience asset. Intact forests regulate water flows, reduce flooding and landslides, cool local climates and sustain livelihoods.

Rewetting drained peatlands — by blocking drainage canals — reduces fire risk and carbon emissions. Community-based forest management, including social forestry permits (*perhutanan sosial*) that give communities legal rights to manage forests, can align conservation with local livelihoods when backed by secure tenure and support.

Planned relocation

In some places, the risks will become too great for people to stay safely. Coastal villages being swallowed by the sea, settlements on unstable slopes, and floodplains that flood repeatedly may eventually need to be relocated. Planned relocation is one of the most difficult forms of adaptation: it severs people from land, livelihoods, graves, community and identity. Experience worldwide shows that it succeeds only when communities lead the decision, when new sites offer livelihoods and services, when social networks are kept together, and when the process is adequately funded. Forced or poorly planned relocation can leave people worse off than before.

KEY POINTS

- Urban resilience requires risk-based planning, space for water, cooling and upgraded informal settlements.
- Stopping groundwater extraction is the key to halting subsidence in coastal cities.
- Mangroves protect coasts, support fisheries and store carbon; restoration must be done carefully.
- Rural resilience rests on water, diverse farming, healthy soils, information, finance and village governance.
- Planned relocation may become necessary; it must be community-led and well resourced.

CHAPTER 15

Technology for Resilience: From Solar Panels to Artificial Intelligence

Technology will not solve the climate crisis by itself, but it can make a profound difference — from the solar lamp that lights a house after a storm to the satellite that warns a village of a coming flood.

The renewable revolution

Over the past fifteen years, the cost of renewable electricity has fallen faster than almost anyone predicted. According to the International Renewable Energy Agency, the global average cost of electricity from new utility-scale solar power plants fell by about 90 percent between 2010 and 2023, and onshore wind by around 70 percent (IRENA, 2024). In most of the world, new solar and wind power is now the cheapest source of new electricity. Battery costs have fallen similarly, making it increasingly practical to store solar energy for use at night.

For Indonesia, which receives abundant sunshine year-round and has vast geothermal, hydro and wind resources, these trends offer an opportunity to meet growing electricity demand while reducing dependence on coal, which still generates around two-thirds of the country's power. Indonesia has some of the world's largest geothermal resources, thanks to its volcanic geology. Floating solar on reservoirs, such as the Cirata floating solar plant in West Java — among the largest in Southeast Asia when it opened in 2023 — shows the potential.

Renewable energy also strengthens resilience. Decentralised systems — rooftop solar with batteries, village mini-grids — can keep lights,

phones, refrigerators for medicines and water pumps running when the central grid fails. For remote islands that rely on expensive diesel fuel, solar mini-grids reduce costs and dependence on fuel shipments that storms can interrupt.

Early warning systems

Early warning systems are among the most cost-effective adaptation measures. As noted in Chapter 8, just 24 hours' warning of a storm or heat wave can reduce damage by 30 percent. The United Nations' *Early Warnings for All* initiative, launched in 2022, aims to ensure that everyone on Earth is protected by early warning systems by 2027.

An effective early warning system has four elements:

1. **Risk knowledge** — understanding the hazards and who is exposed and vulnerable.
2. **Monitoring and forecasting** — observing conditions and predicting hazards.
3. **Warning dissemination and communication** — getting clear, timely, actionable warnings to everyone at risk, including the most remote and marginalised.
4. **Preparedness and response capability** — ensuring people know what to do when they receive a warning, and are able to do it.

Systems often fail at the "last mile": warnings are generated but do not reach people, are not understood, or are not trusted or acted upon. This is why community involvement, multiple communication channels (sirens, mosques, radio, SMS, apps, messengers on motorbikes), and regular drills are so important.

Indonesia operates several warning systems: BMKG's weather and climate warnings and the Indonesia Tsunami Early Warning System (InaTEWS), established after the 2004 tsunami; volcano monitoring by PVMBG; flood warnings in some river basins; and fire danger ratings. The 2018 Sulawesi earthquake and tsunami and the Sunda Strait tsunami of the same year — caused by a volcanic flank collapse that the standard earthquake-based system could not detect — revealed gaps that have since driven improvements. Each event underscores that technology must be combined with public awareness, so that people respond to natural warning signs even when official alerts fail.

Data, satellites and artificial intelligence

The amount of data about the Earth's climate and environment has exploded. Satellites monitor temperature, rainfall, sea level, vegetation, fires, floods and land subsidence in near real time. Ground sensors, weather stations and even mobile phones provide additional data.

Artificial intelligence (AI) and machine learning are being applied to this data in many ways:

- **Weather forecasting.** AI-based models developed in recent years can produce global weather forecasts that rival or exceed traditional physics-based models in accuracy for many variables, at a fraction of the computing cost. National meteorological agencies are beginning to integrate them alongside traditional models.
- **Flood forecasting.** Machine-learning models trained on river and rainfall data can extend flood warnings to river basins that lack dense monitoring networks, and some services already provide flood forecasts for many countries.
- **Fire detection.** Satellite data processed with algorithms detects hotspots, helping authorities respond to fires early. Indonesia uses satellite hotspot data for fire monitoring.
- **Disease surveillance.** Models that combine climate data with disease reports can predict dengue outbreaks weeks in advance, giving health authorities time to act.
- **Damage assessment.** After disasters, AI analysis of satellite and drone images can rapidly map damaged buildings and roads, guiding relief efforts.
- **Agriculture.** Apps provide farmers with tailored advice on planting, irrigation and pest control based on weather forecasts and soil data.

AI is not a panacea. Models are only as good as the data they are trained on, and data are often sparse in exactly the places that are most vulnerable. AI systems can be opaque, biased and misused, and data centres consume significant energy and water. The most useful applications put AI in service of human decision-makers and communities, with transparency about uncertainties.

> **USING TECHNOLOGY WISELY**
>
> The best technology for resilience is often the simplest one that works reliably in local conditions. A battery radio may be more useful than an app when the mobile network is down. A well-maintained hand pump

> may serve a village better than an advanced treatment plant without spare parts or trained operators. Ask: Does it work without electricity or internet? Can it be repaired locally? Is it affordable? Do people trust it and know how to use it?

Low-tech and appropriate technology

Not all resilience technology is high-tech. Many of the most effective tools are simple, cheap and locally made:

- ceramic water filters and solar water disinfection;
- rainwater harvesting tanks and infiltration wells;
- improved cookstoves that reduce smoke and fuel use;
- evaporative coolers (clay pot refrigerators) to keep vegetables fresh;
- hand-crank radios and solar lanterns;
- bamboo structures for coastal protection and earthquake-resistant houses;
- traditional *kentongan* (slit drums) for local warnings.

Appropriate technology is suited to local conditions, affordable, maintainable with local skills and materials, and acceptable to the people who use it.

Digital tools for households and communities

Smartphones put useful resilience tools in many hands:

- **Info BMKG** — weather forecasts, extreme weather warnings, earthquake notifications.
- **InaRISK Personal** — hazard information for your location and preparedness guidance.
- **Messaging groups** — neighbourhood and family groups for rapid communication.
- **Mobile banking and e-wallets** — quick access to funds and aid transfers.
- **Cloud storage** — secure copies of important documents.
- **Offline maps** — navigation when networks are down.

Digital tools also carry risks: misinformation spreads quickly on social media during disasters (Chapter 18), and not everyone has a smartphone or the skills to use it. Communities should ensure that information reaches people through multiple channels.

Technology and mitigation

Technology is essential not only for adapting to climate change but for limiting it. Beyond renewable energy, key technologies include electric vehicles and two-wheelers (particularly relevant in Indonesia, with its tens of millions of motorbikes), energy-efficient appliances and buildings, low-emission cement and steel, methane reduction in agriculture and waste, and, more controversially, carbon capture and storage. Indonesia's nickel reserves, a key ingredient for batteries, place it at the centre of the global electric vehicle supply chain — an opportunity that must be balanced against the environmental and social impacts of mining and smelting.

KEY POINTS

- Solar power costs fell about 90 percent from 2010 to 2023; renewables are now the cheapest new power.
- Decentralised solar and batteries keep essential services running during outages.
- Early warning systems save lives when they reach the last mile and people know how to act.
- AI improves forecasting, flood and fire detection and disease prediction, but has limits.
- Simple, appropriate technologies are often the most reliable.

CHAPTER 16

The Mind in Crisis: Anxiety, Trauma and Psychological Resilience

Climate change affects not only bodies, homes and livelihoods but minds. Fear, grief, trauma and exhaustion are natural responses to a threatening situation. Learning to recognise and manage them is part of surviving the future.

Climate anxiety

Many people, especially the young, feel worry, fear, anger, sadness, guilt or helplessness when they think about climate change. A 2021 survey of 10,000 young people aged 16 to 25 in ten countries found that 59 percent were very or extremely worried about climate change, and more than 45 percent said their feelings about climate change negatively affected their daily lives and functioning. Many felt betrayed by governments' inaction (Hickman et al., 2021).

Climate anxiety is not a mental illness. It is a rational emotional response to a real threat, and it can motivate action. But when it becomes overwhelming — when it disrupts sleep, concentration, relationships or the ability to function — it needs attention.

Helpful strategies include:

- **Acknowledge feelings.** Naming emotions and sharing them with others reduces their intensity. You are not alone.
- **Balance information.** Stay informed, but limit exposure to distressing news, especially before sleep. Seek out stories of solutions and progress as well as problems.
- **Take action.** Action is one of the most effective antidotes to anxiety. Choose actions within your control — preparing your household,

joining a community group, changing habits, supporting good policies. Collective action is particularly powerful because it builds connection as well as impact.
- **Connect with nature.** Spending time in green spaces, gardens, parks or by the sea supports mental well-being.
- **Look after the basics.** Sleep, physical activity, healthy food and time with loved ones build resilience to stress.
- **Seek help when needed.** If anxiety becomes overwhelming, talk to a health professional.

Trauma after disasters

Disasters can be deeply traumatic. Survivors may have faced the threat of death, lost loved ones, homes and possessions, witnessed terrible events or been separated from family. In the days and weeks afterwards, most people experience some distress: shock, fear, sadness, anger, guilt, difficulty sleeping, nightmares, being easily startled, or feeling numb. These are normal reactions to abnormal events, and for most people they gradually fade.

For a minority, distress persists or worsens, developing into conditions such as post-traumatic stress disorder (PTSD), depression, anxiety disorders or substance misuse. Signs that professional help may be needed include symptoms lasting more than a month, severe difficulty functioning at home, school or work, thoughts of self-harm or suicide, and heavy use of alcohol or drugs.

Children and young people

Children are particularly vulnerable after disasters, but they are also remarkably resilient when they receive support. They may react by becoming clingy, regressing to earlier behaviours (bedwetting, thumb-sucking), having nightmares, being irritable or withdrawn, or re-enacting the disaster in play.

Adults can help by:
- staying calm and reassuring children that they are safe and that adults are taking care of them;
- restoring routines — meals, sleep, school, play — as quickly as possible;
- listening and answering questions honestly, in age-appropriate language;

- limiting exposure to disturbing images on television and social media;
- giving children simple, meaningful tasks to help, which restores a sense of control;
- watching for signs that a child needs extra support.

Schools play a crucial role in recovery, providing routine, social contact, safe spaces and access to psychosocial support.

Psychological first aid

Psychological first aid (PFA) is a humane, supportive response to people who are suffering after a crisis. It is not professional counselling; anyone can learn it. The World Health Organization's guide summarises PFA in three action principles: **Look, Listen and Link** (WHO, 2011).

- **Look** — check for safety, identify people with obvious urgent basic needs, and notice people showing signs of serious distress.
- **Listen** — approach people who may need support, ask about their needs and concerns, listen without pressuring them to talk, and help them feel calm.
- **Link** — help people address basic needs and access services, help them cope with problems, give accurate information, and connect them with loved ones and social support.

PFA emphasises what *not* to do as well: do not force people to talk about what happened, do not make promises you cannot keep, do not give false information, and do not judge people's reactions.

INDONESIA FOCUS: PSYCHOSOCIAL SUPPORT AFTER DISASTERS

After major disasters in Indonesia — the 2004 Aceh tsunami, the 2006 Yogyakarta earthquake, the 2018 Lombok earthquakes and the 2018 Central Sulawesi earthquake, tsunami and liquefaction — psychosocial support programmes involving health workers, psychologists, teachers, religious leaders and volunteers have helped communities recover. Religious faith, community rituals and the support of extended families are important sources of resilience for many Indonesians. The Ministry of Health, the Indonesian Psychological Association (HIMPSI), PMI and many NGOs provide psychosocial support services, and community health centres (*puskesmas*) can refer people to mental health care.

Building psychological resilience

Psychological resilience is the ability to adapt well to adversity, trauma and stress. It is not a fixed trait but a set of skills and resources that can be developed. Research on resilience after disasters highlights several factors (Hobfoll et al., 2007):

1. **A sense of safety** — physical safety and the feeling of being safe.
2. **Calming** — strategies to reduce physiological arousal, such as breathing exercises, relaxation and routine.
3. **Self- and community efficacy** — the belief that one's actions, individually and collectively, can make a difference.
4. **Connectedness** — relationships with family, friends and community.
5. **Hope** — the belief that things will work out, grounded in realistic expectations.

Preparation itself builds resilience. People who have planned and practised for emergencies feel more in control and respond more effectively, which in turn reduces trauma.

Supporting the supporters

Health workers, rescuers, volunteers, community leaders and family caregivers carry heavy burdens during and after disasters. They are at risk of burnout, secondary trauma and exhaustion. Organisations should provide rest, rotation, debriefing and access to support; individuals should recognise their limits, take breaks and seek help. Looking after oneself is not selfish; it is necessary to keep helping others.

From despair to purpose

There is a difference between optimism (expecting things to turn out well) and hope (believing that one's actions can make a difference, even in difficult circumstances). The climate crisis calls for hope of the second kind — "active hope" grounded in reality and expressed through action. Many people who work on climate change find that engagement, community and purpose transform their anxiety into determination.

> **TAKE ACTION: A BREATHING EXERCISE FOR STRESSFUL MOMENTS**
> Sit comfortably. Breathe in slowly through your nose for a count of four. Hold for a count of four. Breathe out slowly through your mouth for a

count of six. Repeat five to ten times. Slow breathing activates the body's relaxation response and can help in moments of acute stress — before an evacuation, after receiving bad news, or when anxiety rises at night.

KEY POINTS

- Climate anxiety is a rational response; action, connection and balanced information help.
- Most people recover from disaster distress; some need professional help.
- Children need safety, routine, honest information and meaningful tasks.
- Psychological first aid: Look, Listen, Link.
- Safety, calm, efficacy, connectedness and hope build resilience.

PART V

Changing the Trajectory

CHAPTER 17

Policy, Finance and Justice

Households and communities can do a great deal, but the scale of the climate crisis demands action by governments, businesses and the international community. Understanding how climate policy works helps citizens hold leaders to account.

The international framework

The modern international climate regime rests on three agreements. The **United Nations Framework Convention on Climate Change** (UNFCCC), adopted in 1992, set the goal of preventing "dangerous anthropogenic interference with the climate system" and established the principle of "common but differentiated responsibilities and respective capabilities" — recognising that rich countries, which emitted most historical greenhouse gases, should lead. The **Kyoto Protocol** of 1997 set binding emission targets for developed countries but did not cover developing countries and lacked participation from the largest emitter of the time, the United States.

The **Paris Agreement** of 2015 took a different approach. Nearly every country agreed to the goal of holding global warming "well below 2 °C" above pre-industrial levels and "pursuing efforts" to limit it to 1.5 °C. Rather than imposing targets from above, it asks each country to set its own Nationally Determined Contribution (NDC), to report transparently on progress, and to strengthen its NDC every five years. A "global stocktake" assesses collective progress; the first, concluded at COP28 in Dubai in 2023, called on countries to transition away from fossil fuels in energy systems and to triple renewable energy capacity by 2030.

The Paris approach has achieved real progress: projected warming by 2100 has fallen substantially since 2015. But the sum of current NDCs

still puts the world on track for warming well above 2 °C, and implementation lags behind pledges.

Adaptation and loss and damage

For many years, international negotiations focused mainly on cutting emissions. As impacts have grown, adaptation and "loss and damage" — impacts that cannot be avoided through mitigation or adaptation — have moved to the centre. At COP27 in 2022, countries agreed to establish a fund for responding to loss and damage, which was operationalised at COP28 in 2023 with initial pledges of several hundred million US dollars. This is a small fraction of the estimated needs of vulnerable countries, which run into hundreds of billions of dollars per year, but the fund represents a historic recognition of responsibility.

Climate finance

Developing countries need large sums to adapt to climate change and transition to clean energy. In 2009, developed countries pledged to mobilise US$100 billion a year in climate finance for developing countries by 2020; the goal was met only in 2022, two years late. At COP29 in Baku in 2024, countries agreed on a new collective goal: developed countries should take the lead in mobilising at least US$300 billion a year by 2035, within a broader aim of scaling up finance from all sources to US$1.3 trillion a year. Many developing countries criticised the target as insufficient.

Much climate finance takes the form of loans rather than grants, adding to the debt burdens of vulnerable countries. Adaptation receives a much smaller share than mitigation, even though the needs are vast. The quality, accessibility and fairness of climate finance are therefore as important as its quantity.

Indonesia's climate policy

Indonesia's approach to climate policy reflects its dual position as a highly vulnerable country and a major emitter with large development needs.

Emission targets. As described in Chapter 3, Indonesia's Second NDC (2025) projects emissions peaking in 2030 at about 1.34–1.49 billion tonnes of CO_2-equivalent and aims for net-zero emissions by 2060 or

sooner. Forests and land use (FOLU) are central: the "FOLU Net Sink 2030" policy aims for the forestry and land-use sector to absorb more carbon than it emits by 2030.

Energy transition. Coal remains dominant in electricity generation. The Just Energy Transition Partnership (JETP), launched in 2022 with an initial pledge of US$20 billion from developed countries and international banks, aims to accelerate the shift to renewables and the early retirement of coal plants. Implementation has been slow: by early 2025 only around US$1.1 billion had been disbursed, and the United States withdrew from the partnership in 2025, though Germany and Japan have continued to lead and total commitments have been reported at more than US$21 billion. Captive coal plants built to power industrial parks, especially for nickel processing, have added to emissions.

Carbon pricing. Indonesia's 2021 Harmonisation of Tax Regulations Law (*UU HPP*) introduced a legal basis for a carbon tax, though its full implementation has been delayed. An emissions trading system for coal-fired power plants began in 2023, and a carbon exchange (IDXCarbon) was launched in September 2023.

Adaptation and disaster management. Indonesia's disaster management law (Law No. 24 of 2007) established BNPB and regional BPBDs and shifted the focus from response to risk reduction. National development plans increasingly incorporate climate resilience, and the government has developed disaster risk financing mechanisms (Chapter 12).

How carbon pricing works

Carbon pricing puts a price on greenhouse gas emissions, making polluters pay for some of the damage they cause and creating an incentive to reduce emissions. There are two main approaches. A **carbon tax** sets a price per tonne of emissions and lets the market determine how much emissions fall. An **emissions trading system** (cap-and-trade) sets a limit on total emissions, distributes or auctions permits, and lets companies trade them, so the market determines the price.

Carbon pricing is considered by many economists to be an efficient way to reduce emissions, but it faces political challenges, particularly concerns about rising energy prices for poor households. Revenues can be used to address these concerns — through rebates, social protection or investment in clean energy and public transport. Fossil fuel subsidies, which in many countries including Indonesia have historically

consumed large shares of public budgets, work in the opposite direction, effectively paying for emissions.

Climate justice

Climate change raises profound questions of justice. Those who contributed least to the problem often suffer most. The richest 10 percent of the world's population is responsible for roughly half of global consumption-based emissions, while the poorest half accounts for around a tenth. Low-income countries, small island states and poor communities within every country face the greatest risks and have the fewest resources to adapt.

Justice also has a temporal dimension: today's emissions impose costs on future generations who have no voice in current decisions. And it has a procedural dimension: those affected by climate policies — including indigenous peoples, women, workers in fossil fuel industries, and communities near mines or dams — deserve a voice in decisions.

A "just transition" means ensuring that the shift to a low-carbon economy is fair: protecting workers and communities dependent on fossil fuels through retraining, social protection and new economic opportunities; respecting the rights of communities affected by new projects such as mines, dams and plantations; and sharing the benefits of clean energy widely.

The role of citizens

Policy does not change by itself. Citizens influence climate policy through elections, public consultation, advocacy, the media, consumer choices, and participation in community and civil society organisations. In Indonesia, local governments have significant authority over spatial planning, disaster management, waste, water and public transport; engaging with them can produce tangible results. Courts are another avenue: climate litigation is growing worldwide, and in 2025 the International Court of Justice issued an advisory opinion affirming that states have legal obligations under international law to protect the climate system.

TAKE ACTION: ENGAGING WITH CLIMATE POLICY

- Learn your local government's disaster management and climate plans, and attend public consultations (*Musrenbang*).
- Ask candidates in elections about their plans for climate adaptation, clean energy and disaster preparedness.
- Support organisations working on climate, environment and disaster risk reduction.
- Use your voice as a consumer, employee and investor to encourage businesses to reduce emissions and risks.
- Join or form community groups that advocate for local adaptation measures — drainage, green space, clean water.

KEY POINTS

- The Paris Agreement asks every country to set and strengthen NDCs; current pledges are not enough.
- Loss and damage and adaptation finance are growing priorities, but funding falls far short of needs.
- Indonesia aims for an emissions peak in 2030 and net zero by 2060; the energy transition is the biggest challenge.
- Carbon pricing can cut emissions efficiently if designed fairly.
- Climate justice and a just transition are essential to legitimacy and success.

CHAPTER 18

Thinking Clearly: Evidence, Misinformation and Climate Literacy

In an age of information overload, the ability to distinguish reliable knowledge from rumour, spin and falsehood is a survival skill. During disasters, it can be a matter of life and death.

Why climate literacy matters

Climate literacy means understanding the essential principles of the climate system, knowing how to assess scientific information, communicating about climate in a meaningful way, and being able to make informed and responsible decisions. It does not require a degree in science. It requires curiosity, a basic understanding of how science works, and habits of critical thinking.

Climate literacy helps individuals make better decisions about where to live, how to prepare, what to buy and whom to vote for. It helps communities resist misinformation. And it helps societies sustain support for policies whose benefits may take years to appear.

How science works

Science is not a collection of fixed facts but a method for reducing uncertainty through evidence. Its key features include:

- **Evidence and testing.** Scientific claims must be testable against observations. Ideas that fail tests are revised or abandoned.

- **Peer review.** Before publication in reputable journals, research is evaluated by independent experts. Peer review is not perfect, but it filters out many errors.
- **Replication and convergence.** Confidence grows when independent studies using different methods reach similar conclusions. The evidence for human-caused climate change rests on many independent lines of evidence — thermometers, satellites, ocean buoys, ice cores, physics and models — that converge.
- **Assessment.** Bodies such as the IPCC do not conduct new research but assess thousands of studies, weighing the evidence and expressing confidence levels. IPCC reports are reviewed by experts and governments and represent the most comprehensive synthesis of climate knowledge.
- **Self-correction.** Science changes as evidence accumulates. This is a strength, not a weakness.

Reading a scientific claim

When you encounter a claim about climate change — in a news article, a social media post, a speech — ask:

1. **Who is making the claim, and what are their qualifications and interests?** Is the source a recognised scientific body, a peer-reviewed journal, a reputable news organisation, or an anonymous account? Does the source have a financial or political interest?
2. **What is the evidence?** Is the claim based on data and research, or on anecdote and opinion? Can you find the original study?
3. **Is it consistent with the broader body of evidence?** A single study that contradicts many others should be treated with caution until it is confirmed.
4. **Is it being presented accurately?** Headlines often exaggerate or oversimplify. Check whether the original study actually supports the claim.
5. **What is the uncertainty?** Good science reports uncertainty. Claims of absolute certainty — or claims that uncertainty means we know nothing — are red flags.

Understanding numbers

Climate information is full of numbers. A few concepts help interpret them:

- **Averages and extremes.** A small change in the average can produce a large change in the frequency of extremes. If the average temperature shifts by 1 °C, days that used to be rare may become common.
- **Trends and variability.** Climate is defined over decades. A single cold day or year does not disprove warming, just as a single hot day does not prove it. Look at long-term trends.
- **Baselines.** Always check what a change is measured against: pre-industrial (1850–1900), a recent period (for example 1991–2020), or the previous year.
- **Relative and absolute risk.** "Doubles the risk" sounds alarming, but if the original risk was 1 in a million, it is now 2 in a million. Conversely, a small percentage change applied to a large population can affect millions.
- **Correlation and causation.** Two things happening together does not mean one causes the other. Attribution science uses careful methods to establish causal links.
- **Likelihood language.** The IPCC uses calibrated terms: "virtually certain" (99–100 percent probability), "very likely" (90–100 percent), "likely" (66–100 percent), "about as likely as not" (33–66 percent), and so on.

Misinformation and disinformation

Misinformation is false or misleading information shared without intent to deceive. **Disinformation** is false information deliberately spread to deceive. Both are common in climate discussions.

Climate misinformation has evolved. Early forms denied that warming was happening or that humans were responsible. As the evidence became overwhelming, misinformation shifted towards claims that impacts will be mild or beneficial, that solutions will not work or are too costly, that climate science is unreliable, or that action is pointless because others are not acting. Researchers have catalogued common techniques, often summarised by the acronym FLICC: **F**ake experts, **L**ogical fallacies, **I**mpossible expectations, **C**herry-picking and **C**onspiracy theories (Cook, 2020).

During disasters, misinformation can be dangerous: false reports of an impending tsunami or eruption can cause panic; false "all clear" messages can lead people to return too early; and rumours about aid distribution can cause conflict. After the 2018 Central Sulawesi earth-

quake and tsunami, hoaxes spread rapidly on social media, prompting authorities to issue repeated clarifications.

Protecting yourself and others

Research shows that people can be "inoculated" against misinformation by learning about manipulation techniques in advance — a strategy called prebunking (van der Linden et al., 2017). Other practical steps include:

- **Pause before sharing.** Emotional content — fear, outrage — spreads fastest. Take a moment to check before forwarding.
- **Check the source.** Is it an official agency (BMKG, BNPB, BPBD, the Ministry of Health), a reputable news outlet, or an unknown account?
- **Look for confirmation.** Do multiple reliable sources report the same information?
- **Use fact-checking services.** Indonesian fact-checkers such as Cek Fakta, the government's anti-hoax portal and news organisations' fact-check sections regularly debunk false claims.
- **Correct gently.** When friends or family share misinformation, respond respectfully with accurate information and a reliable source. Ridicule tends to entrench beliefs.
- **During disasters, rely on official channels.** Follow BMKG, BNPB and your local BPBD for warnings and instructions.

INDONESIA FOCUS: OFFICIAL SOURCES

- **BMKG** — bmkg.go.id; *Info BMKG* app; official social media accounts — weather, climate, earthquakes, tsunamis, air quality.
- **BNPB** — bnpb.go.id; InaRISK; official social media — disaster information, risk maps, situation reports.
- **PVMBG / MAGMA Indonesia** — magma.esdm.go.id — volcanic activity and alert levels.
- **Ministry of Health** — kemkes.go.id — disease outbreaks and health guidance.
- **Local BPBD** — regional warnings, evacuation orders and shelters.

Talking about climate change

Many people want to discuss climate change with family, friends or colleagues but are unsure how. Research on climate communication suggests:

- **Start with shared values** — family, faith, health, community, prosperity — rather than with facts alone.
- **Make it local and personal.** Talk about how climate change affects the places and things people care about: the flooding on our street, the heat in our children's school, the price of rice.
- **Focus on solutions.** People engage more when they see that action is possible and effective.
- **Listen.** Understanding others' concerns builds trust and opens minds more than lecturing.
- **Use trusted messengers.** Religious leaders, doctors, teachers, farmers and community figures are often more persuasive than distant experts.

KEY POINTS

- Climate literacy means understanding the basics, evaluating evidence and making informed decisions.
- Science works through testing, peer review, replication and self-correction.
- Ask who, what evidence, consistent, accurate, and how uncertain.
- Misinformation uses fake experts, logical fallacies, impossible expectations, cherry-picking and conspiracy theories.
- Pause before sharing; rely on official channels during disasters.

CHAPTER 19

A Life Plan for a Changing Century

Surviving the future is not a single act but a way of living: informed, prepared, connected and engaged. This final chapter brings the book's lessons together into a practical plan for the years ahead.

Looking back, looking forward

This book has covered a great deal of ground: the physics of the greenhouse effect and the evidence of warming; the scenarios that describe possible futures; the particular exposure of Indonesia and Southeast Asia; the hazards of heat, water, food insecurity, disease and cascading crises; the practical steps of household preparedness; the power of communities, adaptive cities and villages, technology and psychological resilience; and the policies, finance and justice that shape the larger trajectory.

A few themes run through all of it.

The warming is real, human-caused and continuing. There is no serious scientific doubt. The coming decades will be hotter than the past, and sea level will keep rising for centuries.

How much worse it gets is still up to us. The difference between 1.5 °C, 2 °C and 3 °C is enormous. Every fraction of a degree avoided reduces suffering.

Preparation saves lives. From *smong* on Simeulue to heat action plans in Ahmedabad and early warning systems around the world, knowledge acted upon reduces death and loss dramatically.

Resilience is collective. No household is resilient alone. Strong communities, competent institutions and fair policies are the deepest foundations of safety.

Justice matters. Those who did least to cause the crisis are suffering most. A response that ignores fairness will be neither legitimate nor effective.

A personal resilience plan

The following plan organises the book's practical recommendations by time horizon. Adapt it to your circumstances.

This week

- Install the *Info BMKG* app and enable warnings for your area.
- Look up your home on InaRISK.
- Save emergency numbers in your phone and write them on paper: 112 (general emergency), 110 (police), 113 (fire), 115 (search and rescue), 117 (BNPB), 118/119 (ambulance).
- Talk with your household about what you would do in a flood, earthquake or heat wave.
- Do a *3M Plus* check around your home to remove mosquito breeding sites.

This month

- Complete a household risk assessment (Chapter 9).
- Write a family emergency plan with meeting points, routes and an out-of-area contact (Chapter 11).
- Pack an emergency bag for each household member.
- Begin storing drinking water and a three-day supply of non-perishable food (Chapter 10).
- Scan important documents and store them securely in the cloud (Chapter 12).
- Open or designate an emergency savings account and make the first deposit.

This year

- Take a first-aid course, ideally including CPR.
- Make at least one heat-proofing and one flood-proofing improvement to your home.
- Review your insurance coverage: health (BPJS), property and, if relevant, crops.
- Build your water and food stock towards two weeks if you live in a high-risk area.
- Get to know your neighbours; join or help start a community preparedness group (Chapter 13).

- Hold a family evacuation drill before the rainy season.
- Reduce your own emissions where you can: energy efficiency, public transport or electric two-wheelers, less food waste, rooftop solar if feasible.

This decade

- Consider climate risk in major decisions: where to buy land or a house, how to build or renovate, what to study, which career or business to pursue.
- Diversify income and assets to reduce exposure to climate shocks.
- Engage with local government on adaptation: drainage, green space, water supply, early warning, mangrove and forest protection.
- Support policies and leaders committed to reducing emissions and protecting the vulnerable.
- Pass on what you know to the next generation — as the people of Simeulue did.

For communities and organisations

Communities, schools, workplaces, religious organisations and businesses can apply the same logic at a larger scale:

- **Assess risks** to people, buildings, operations and supply chains.
- **Plan** for continuity: who does what in an emergency, how to communicate, how to keep essential services running.
- **Invest** in prevention: drainage, cooling, backup power and water, nature-based solutions.
- **Protect the vulnerable**: identify those who need help and assign responsibilities.
- **Practise** through drills and exercises.
- **Learn** from every event and update plans.
- **Reduce emissions** through energy efficiency, renewable energy and sustainable practices.

Reasons for hope

It is easy to feel overwhelmed by the scale of the climate crisis. But there are real grounds for hope — not naive optimism, but hope grounded in evidence.

Renewable energy is now the cheapest source of new electricity in most of the world, and its growth has exceeded expectations year after year. Projected warming by 2100 has fallen significantly since the Paris

Agreement, from more than 4 °C under the old business-as-usual to around 2.5–3 °C under current policies — still dangerous, but proof that action works. Early warning systems have dramatically reduced deaths from cyclones in countries such as Bangladesh and India, where tropical cyclones that once killed hundreds of thousands now kill far fewer. Indonesia itself has built a disaster management system that did not exist two decades ago, and its subsidence in Jakarta has slowed where groundwater use was curbed.

Above all, human beings have an extraordinary capacity for cooperation, innovation and care. *Gotong royong* is not a relic; it is a resource. Every community that cleans its drains, plants mangroves, checks on its elderly during a heat wave, or teaches its children what to do when the ground shakes is writing a different future.

A final word

The future described in this book is demanding. It will test individuals, families, communities and nations. But it is not a future without hope or choice. We know what is happening. We know what works. The question is whether we will act — early enough, fairly enough, and together.

As the IPCC warned in 2022, "any further delay in concerted anticipatory global action on adaptation and mitigation will miss a brief and rapidly closing window of opportunity to secure a liveable and sustainable future for all" (IPCC, 2022). That window is still open. Let us use it.

Appendix A: Preparedness Checklists

These checklists summarise the practical guidance in Parts III and IV. Photocopy them, adapt them to your household, and review them at least once a year — ideally before the rainy season.

A.1 Household risk assessment

- ☐ Home address checked on InaRISK; hazard levels noted
- ☐ *Info BMKG* app installed with notifications enabled
- ☐ Past disasters in the area discussed with older neighbours
- ☐ Household members needing extra help identified
- ☐ Risk matrix completed (likelihood × impact)
- ☐ Top three priority actions chosen, with dates

A.2 Family emergency plan

- ☐ Warning channels identified (app, radio, BPBD, mosque, siren, *kentongan*)
- ☐ Safe place in the home for each hazard
- ☐ Meeting point near home
- ☐ Evacuation destination outside the neighbourhood
- ☐ Two evacuation routes; tsunami route to high ground if coastal
- ☐ Out-of-area contact person agreed
- ☐ Responsibilities assigned (children, older relatives, pets, emergency bag)
- ☐ School and workplace emergency plans known
- ☐ Plan written down and shared with all household members
- ☐ Drill held (date: _____)

A.3 Emergency bag (per person or per household)

- ☐ Drinking water (1–2 litres per person) and ready-to-eat food
- ☐ Copies of documents in a waterproof bag; digital copies in the cloud
- ☐ Cash in small denominations
- ☐ Flashlight or headlamp, spare batteries, power bank, cables

- ☐ Battery or hand-crank radio; whistle
- ☐ First-aid kit; two weeks of regular medicines; copies of prescriptions
- ☐ Change of clothes, raincoat, sturdy footwear, blanket or sarong
- ☐ N95 masks, soap, hand sanitiser, toothbrush, sanitary pads
- ☐ Items for infants, older people or people with disabilities
- ☐ Paper list of emergency contacts and a copy of the family plan
- ☐ Knife or multi-tool, duct tape, plastic sheet, strong bags
- ☐ Checked and refreshed (date: _____)

A.4 Home supplies (three days minimum, two weeks in high-risk areas)

- ☐ Drinking water: 4 litres per person per day
- ☐ Water treatment: means to boil, filter or chlorinate
- ☐ Non-perishable food, rotated "first in, first out"
- ☐ Manual can opener; stove and fuel for outdoor or safe use
- ☐ Lights and batteries; solar lantern or small solar kit
- ☐ Hygiene supplies; materials for an emergency toilet
- ☐ Fire extinguisher; gas cylinder secured

A.5 Heat-wave readiness

- ☐ Coolest room in the house identified
- ☐ Curtains or shading for sun-facing windows
- ☐ Fans working; ventilation openings unobstructed
- ☐ Oral rehydration salts and thermometer at home
- ☐ Daily check-in arranged for older or vulnerable relatives and neighbours
- ☐ Outdoor work rescheduled to cooler hours where possible
- ☐ Everyone knows the signs of heat stroke and to call 112 or 119

A.6 Flood readiness

- ☐ Drains and gutters cleared before the rainy season
- ☐ Electrical outlets and appliances raised where possible
- ☐ High shelf or upper floor reserved for documents and supplies
- ☐ Main electricity switch location known to all adults
- ☐ Boots and gloves available for clean-up
- ☐ Evacuation trigger agreed (for example, water at a certain level or an official order)

A.7 Financial resilience

- ☐ Separate emergency savings with regular deposits
- ☐ Some cash kept at home
- ☐ All documents scanned and stored securely
- ☐ BPJS Kesehatan active for all family members
- ☐ Property insurance reviewed for flood and earthquake coverage

- ☐ Crop or livestock insurance considered (farmers)
- ☐ Lenders checked for OJK registration

A.8 Community

- ☐ Neighbourhood messaging group or phone tree in place
- ☐ Community risk and resource map drawn
- ☐ Confidential list of residents needing help, with assigned buddies
- ☐ Regular *kerja bakti* for drains and mosquito breeding sites
- ☐ Contact established with local BPBD, village office and *puskesmas*
- ☐ Community drill held (date: _____)

Appendix B: Emergency Contacts and Information Sources (Indonesia)

Numbers and addresses were correct at the time of writing. Check with your local government for regional numbers, and save them in your phone and on paper.

B.1 Emergency numbers

Service	Number
National emergency call centre (all emergencies)	**112**
Police	110
Fire brigade	113
Search and rescue (Basarnas)	115
National Disaster Management Agency (BNPB)	117
Ambulance / medical emergency	118 or 119
Natural disaster command post	129
State electricity company (PLN)	123

The 112 service operates under Ministry of Communication regulations and is run by local governments, which connect it to fire, health, police, BPBD and other services. It can be dialled free of charge from mobile phones.

B.2 Official information sources

Institution	What it provides	Where
BMKG	Weather forecasts, extreme weather warnings, seasonal outlooks, earthquakes, tsunami warnings, air quality	bmkg.go.id; *Info BMKG* app

Institution	What it provides	Where
BNPB	Disaster information, risk maps, situation reports, preparedness guidance	bnpb.go.id; inarisk.bnpb.go.id; *InaRISK Personal* app
BPBD (regional)	Local warnings, evacuation orders, shelters, contingency plans	Provincial and district government websites and social media
PVMBG / MAGMA Indonesia	Volcano alert levels, eruption information, landslide susceptibility	magma.esdm.go.id
Ministry of Health	Disease outbreaks, health guidance, dengue prevention	kemkes.go.id
Ministry of Environment	Air quality, environmental information, *Program Kampung Iklim*	Ministry website
Indonesian Red Cross (PMI)	First-aid training, blood donation, disaster relief	pmi.or.id
OJK	Registered financial institutions and lenders; consumer complaints	ojk.go.id; call centre 157

B.3 International sources for further reading

- **IPCC** (ipcc.ch) — assessment reports on climate science, impacts, adaptation and mitigation; the Summaries for Policymakers are the most accessible entry point.
- **WMO** (wmo.int) — annual *State of the Global Climate* reports and early warning initiatives.
- **The *Lancet* Countdown** (lancetcountdown.org) — annual indicators on health and climate change.
- **Copernicus Climate Change Service** (climate.copernicus.eu) — monthly climate bulletins and data.
- **Indicators of Global Climate Change** (igcc.earth) — annual updates of key climate indicators and the remaining carbon budget.
- **World Health Organization** (who.int) — guidance on heat, health emergencies and psychological first aid.
- **AHA Centre** (ahacentre.org) — ASEAN coordination on disaster management.

B.4 Personal emergency contacts

Contact	Name	Phone
Out-of-area contact		
Nearest relative		
Neighbour		
RT / RW head		
Nearest *puskesmas* / hospital		
Local BPBD		
Family doctor		
Insurance company		
Workplace		
Children's school		

Glossary

Adaptation — Adjustments in human and natural systems in response to actual or expected climate change, to reduce harm or exploit opportunities.

Aerosols — Tiny particles suspended in the air, from sources such as pollution, fires, dust and volcanoes. Many reflect sunlight and partly mask warming.

Attribution science — The study of how much climate change has influenced the likelihood or intensity of particular extreme events.

Carbon budget — The total amount of carbon dioxide that can still be emitted while keeping warming below a given level with a given probability.

Carbon dioxide (CO_2) — The main long-lived greenhouse gas produced by human activities, chiefly the burning of fossil fuels and deforestation.

Carbon pricing — Policies that put a price on greenhouse gas emissions, through a carbon tax or an emissions trading system.

Climate — The average weather and its variability over a long period, typically 30 years or more.

Climate anxiety — Distress, worry or fear related to climate change and its threats.

Climate finance — Funding for actions that reduce emissions or help societies adapt to climate change.

Committed warming — Warming that will occur because of past emissions and the slow response of the climate system, regardless of future emissions.

Coral bleaching — The loss of the symbiotic algae that give corals their colour and much of their food, typically caused by unusually warm water. Prolonged bleaching kills corals.

Disaster risk reduction (DRR) — Policies and practices aimed at preventing new disaster risk, reducing existing risk and managing residual risk.

Early warning system — An integrated system of hazard monitoring, forecasting, risk assessment, communication and preparedness that enables timely action.

El Niño / La Niña — Opposite phases of a natural climate pattern in the tropical Pacific. El Niño typically brings drought to Indonesia; La Niña brings heavier rain.

Exposure — The presence of people, livelihoods, assets and ecosystems in places that could be adversely affected by hazards.

Greenhouse effect — The warming of the Earth's surface by gases in the atmosphere that absorb and re-emit infrared radiation.

Greenhouse gas — A gas that absorbs infrared radiation, such as carbon dioxide, methane, nitrous oxide and water vapour.

Hazard — A potentially damaging physical event, such as a flood, heat wave, landslide or earthquake.

Heat index — A measure combining air temperature and humidity to represent how hot it feels to the human body.

Heat stroke — A life-threatening condition in which body temperature rises above about 40 °C and the body's cooling mechanisms fail; requires immediate cooling and medical care.

Indian Ocean Dipole (IOD) — A natural pattern of differing sea surface temperatures between the western and eastern Indian Ocean that affects rainfall in Indonesia.

IPCC — The Intergovernmental Panel on Climate Change, the United Nations body that assesses climate science.

Land subsidence — The sinking of the ground surface, often caused by excessive groundwater extraction and the compaction of soft soils.

Loss and damage — Climate impacts that cannot be avoided through mitigation or adaptation.

Mitigation — Actions to reduce greenhouse gas emissions or enhance their removal from the atmosphere.

Nationally Determined Contribution (NDC) — A country's climate action plan under the Paris Agreement, including emission targets.

Net-zero emissions — A state in which greenhouse gas emissions released into the atmosphere are balanced by removals.

Paris Agreement — The 2015 international agreement to hold warming well below 2 °C and pursue efforts to limit it to 1.5 °C.

Peatland — Waterlogged land with thick layers of partly decomposed plant material that store large amounts of carbon; highly flammable when drained.

Psychological first aid (PFA) — A humane, supportive response to people suffering after a crisis, summarised as Look, Listen and Link.

Resilience — The capacity of people, communities and systems to anticipate, absorb, recover from and adapt to shocks and stresses.

Shared Socioeconomic Pathways (SSPs) — Scenarios used by the IPCC to describe alternative futures of society, emissions and climate.

Social capital — The networks, norms and trust that enable people to act together.

Storm surge — An abnormal rise of sea level during a storm, caused mainly by strong winds pushing water towards the shore.

Tipping point — A critical threshold beyond which a part of the climate system changes abruptly and possibly irreversibly.

Urban heat island — The phenomenon in which cities are warmer than surrounding rural areas because of buildings, paved surfaces and human activity.

Vulnerability — The propensity or predisposition of people or systems to be adversely affected by hazards.

Wet-bulb temperature — The lowest temperature to which air can be cooled by evaporating water; an indicator of how well sweating can cool the body.

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Index

A
adaptation, 12–13, 19, 35–36, 48–49, 73, 81–82, 85, 87–88, 90
attribution science, 7, 111

B
BMKG, 16, 27, 30, 37, 43, 56, 58–59, 67, 90, 92, 112, 116
BNPB, 19, 30, 56, 81, 105, 112, 116

C
carbon budget, 7–8
carbon pricing, 105, 107
climate anxiety, 95, 99
climate finance, 104
coral reefs, 11–14, 18, 36, 87

D
dengue, 11, 18–20, 41–42, 44, 57, 91
drought, 7, 11, 15, 17–20, 29, 31, 33, 35–38

E
early warning, 16, 20, 26, 48, 50–51, 56, 80–81, 87, 90, 93, 115, 117
El Niño, 6, 15, 17–18, 20, 31, 33, 58
emergency kit, 68
evacuation, 30, 56–57, 67–69, 71, 76, 81–83, 99, 112, 117

F
fires and haze, 17–18, 20, 43–44, 56, 69
flooding, 7, 11, 15–17, 19, 23, 29–30, 32–33, 35–37
food security, 35, 41

G
gotong royong, 80, 83, 118
greenhouse effect, 4, 8, 115
groundwater, 11, 16–17, 20, 31–33, 85, 88, 118

H
heat stroke, 25, 27, 43
heat waves, 7, 18, 26–27, 36, 41, 44, 47–48, 55, 57, 90, 116, 118

I
insurance, 7, 37, 39, 57, 68, 70, 73–76, 87, 116
IPCC, 5–7, 9–11, 13, 29, 32, 47, 110–111, 118

J
Jakarta, 11, 16–17, 19, 86, 118
JETP, 19, 105

L
land subsidence, 11, 14, 16–17, 19–20, 30, 58, 85, 88, 91, 118
loss and damage, 104, 107

M
mangroves, 13, 17, 31, 48, 50, 86–88, 117–118
mental health, 41, 43–44, 97
migration, 19, 47–49, 51
misinformation, 92, 109, 111–113
mutual aid, 79, 81, 83

N
NDC, 19–20, 103–104, 107

P
Paris Agreement, 12, 103, 107, 117
peatlands, 4, 15, 17–20, 38, 43, 56, 87–88
psychological first aid, 97, 99

R
rice, 4, 15, 17, 19, 31, 35–38, 58, 62, 75, 87, 113

S
sea-level rise, 5, 7, 10–14, 16–17, 20, 30, 33, 48, 50
Simeulue, 58, 80, 83, 115, 117
smong, 58, 80, 83, 115
solar power, 6, 50, 62–65, 89–93, 117

T
tipping points, 9, 11–14

U
urban heat island, 25

W
wet-bulb temperature, 24
Wolbachia, 19, 42, 44